

Appendix B

Labeling, Design and Syntax Standards

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Appendix B

Labeling, Design and Syntax Standards

The following information is provided to further define the end results that the Cal Poly ITS Telecomm group shall use to evaluate Division 27 and related work performed on the Cal Poly Campus.

Figure number labeling of various items and locations:



Figure 101: Backboard label to be stenciled on appropriate ER/TR/EF wall (Building 171A, Room 21 – West Wall) in every telecomm space.



Figure 102: Hard-line CATV Cable label between ER and TR (Building 171A-Room 21 to Building 171A-Room 217).



Figure 103: Labeling of a fiber shelf in a rack (Fiber- Multimode-Building 2 to Building 14-Room 103- Multimode strands 1 to 24.

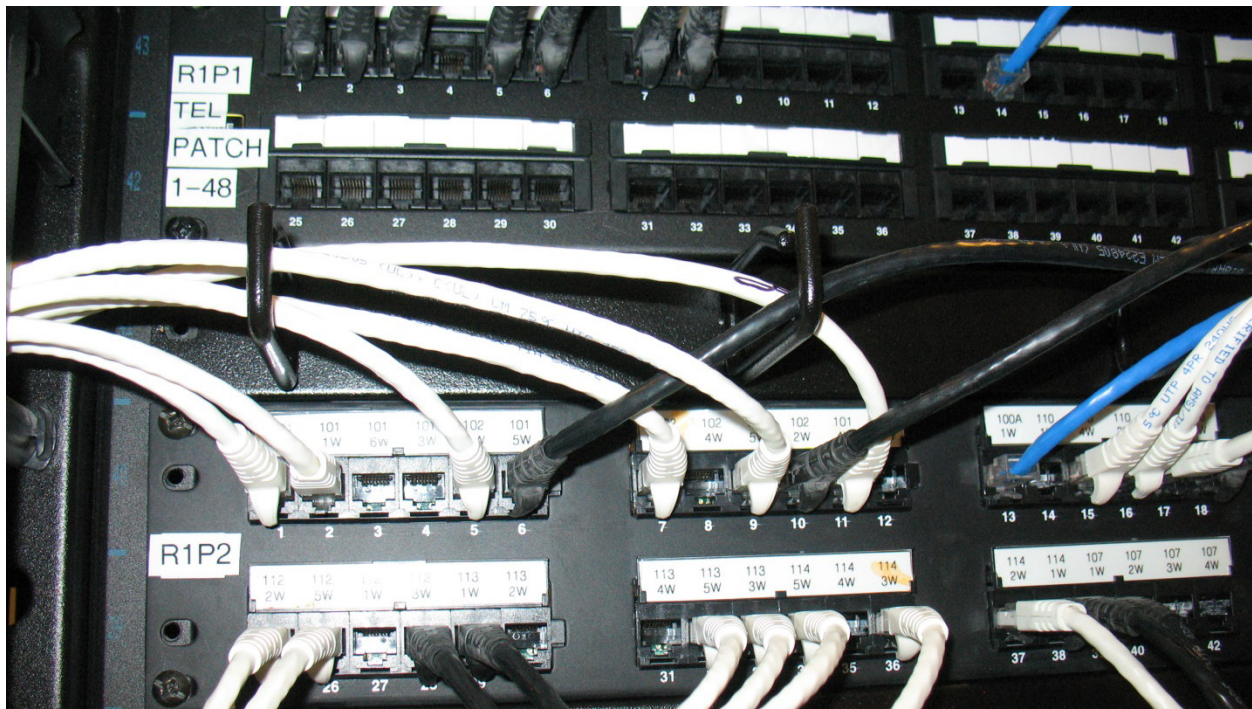


Figure 104: Patch panel labeling, R1P1, 1-48 phone and R1P2 (Rack 1, Patch Panel 2) data.

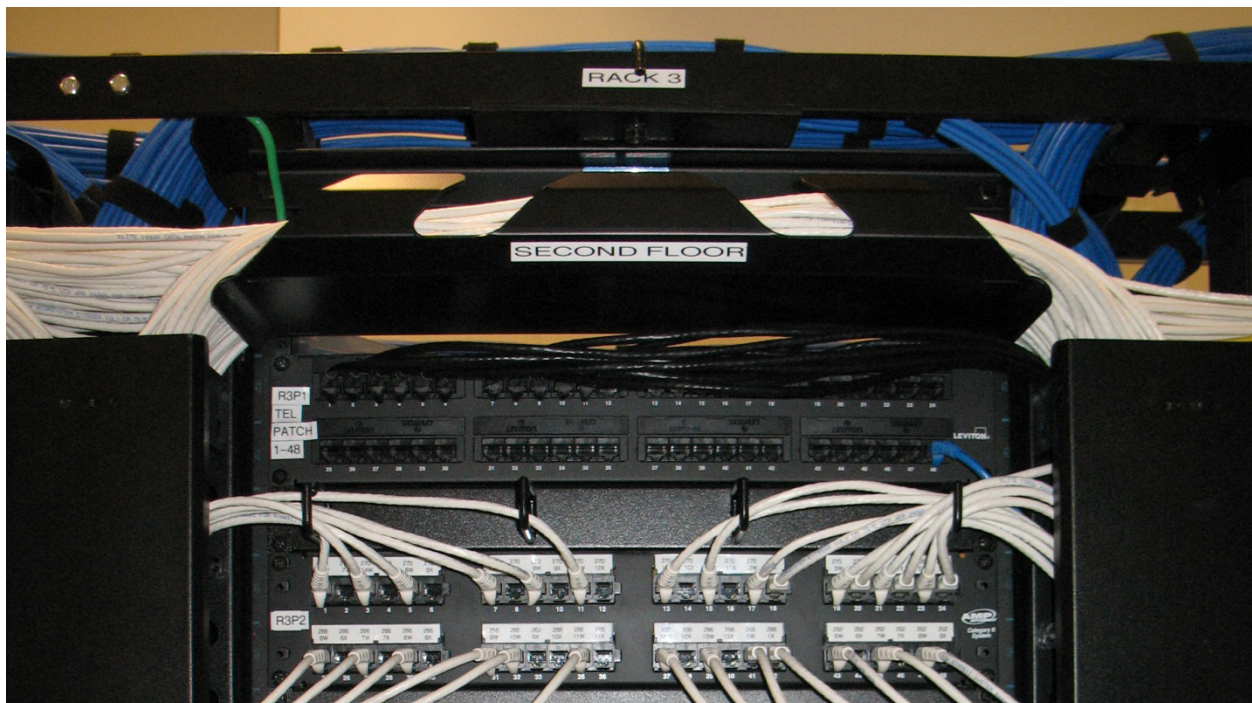


Figure 105: Rack labeling is affixed to the ladder racking above and centered on the rack. Floor location is affixed to the cable trough mounted at the top of each rack.



Figure 106: Protector labeling for a cable pairs 1 through 100 that goes from building 20B, room 100 (only room in building) to building 180, room 106.



Figure 107: Structured Cable, CAT 6, labeled with the faceplate room and port served.



Figure 108: Two 4 port faceplates in the most common configurations found on campus. The upper label identifies the specific faceplate within a room. A number such as 204A-01 would mean, when entering room 204A, which is a room within room 204, starting on your left and going around the room, faceplate 204A-01 would be the first faceplate you should come to in that room. The lower label indicates the room number of the TR/ER that supports the jacks installed in the faceplate. The wiring connected to the rear of the connectors/jacks (see Figure 107) should have labels identifying which port they serve; “W” (upper left), “X” (upper right), “Y” (lower left) or “Z” (lower right).

LABELING STANDARDS FACEPLATES

Use this format to label Faceplates, Campus wide.

FACEPLATES NEED THREE TYPES OF LABELS ON THEM:

- The Top Label indicates the Room and Faceplate location Number.
- The Lower Label indicates the Telecommunications Space the Cables go to.
- Be sure to label the Jacks (WXYZ) properly for Angled and non-Angled Faceplates. In some Angled-style Faceplates, the Jacks are “up-side-down.”

- The U, V, W, X, Y and Z Jacks should always be in the same relative position. If the Faceplate needs to be positioned on its side, turn it 90 degrees counter-clockwise (the “top” now on the left).
- All labels must be mechanically produced.

Examples: Here are some examples of various Faceplate configurations.

Angled 4 Port

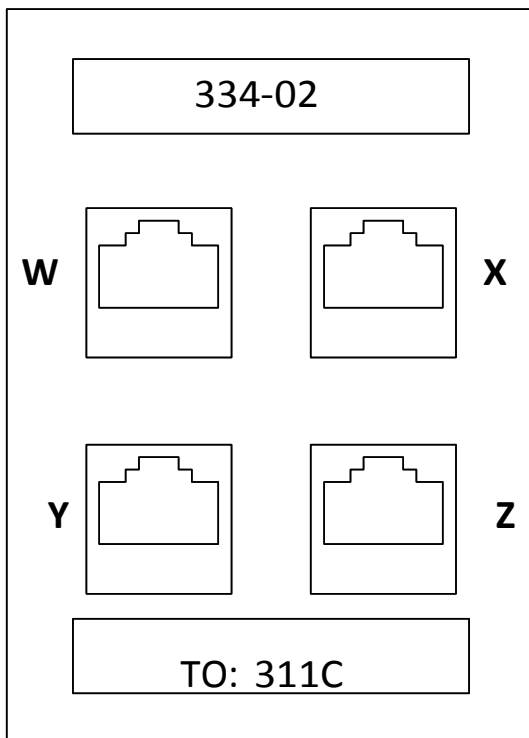


Figure
109

Straight 4 Port

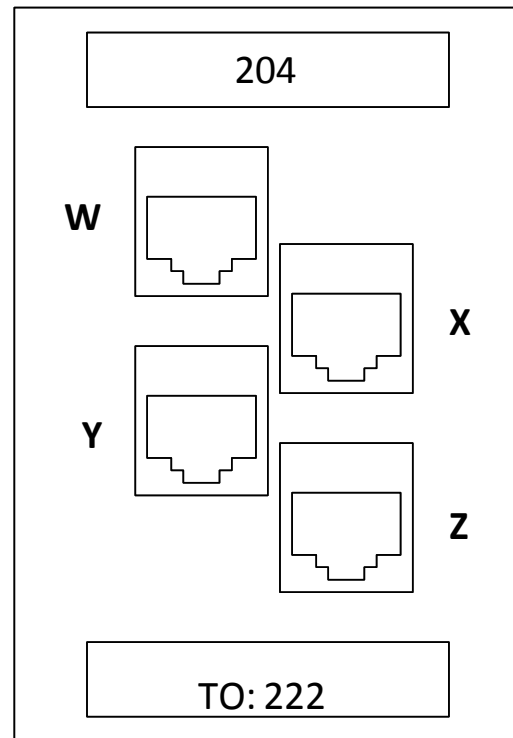


Figure
110

Straight 3 Port with Cable TV

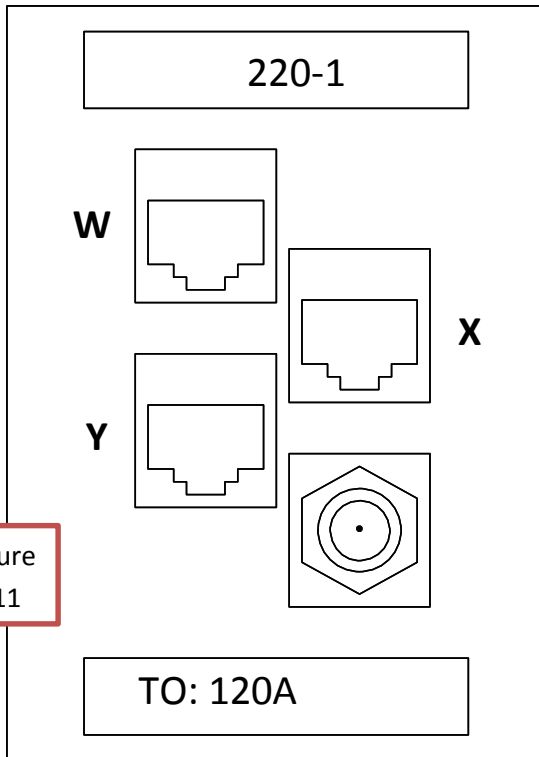


Figure 111

Straight 2 Port

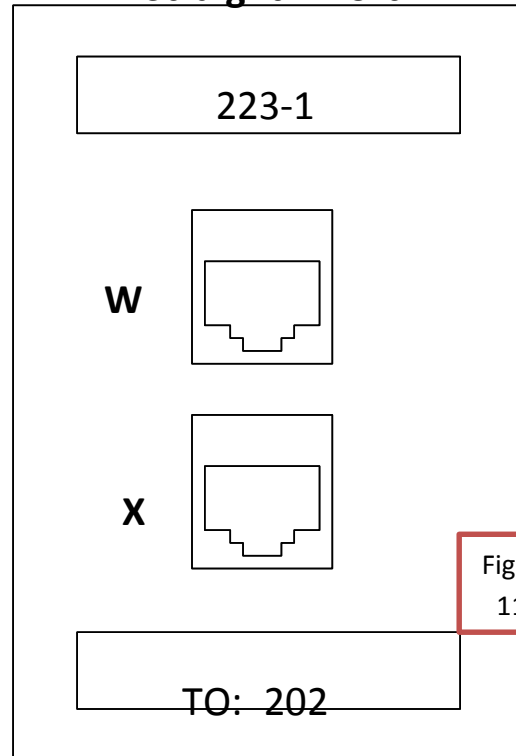


Figure 112

Angled 3 Port

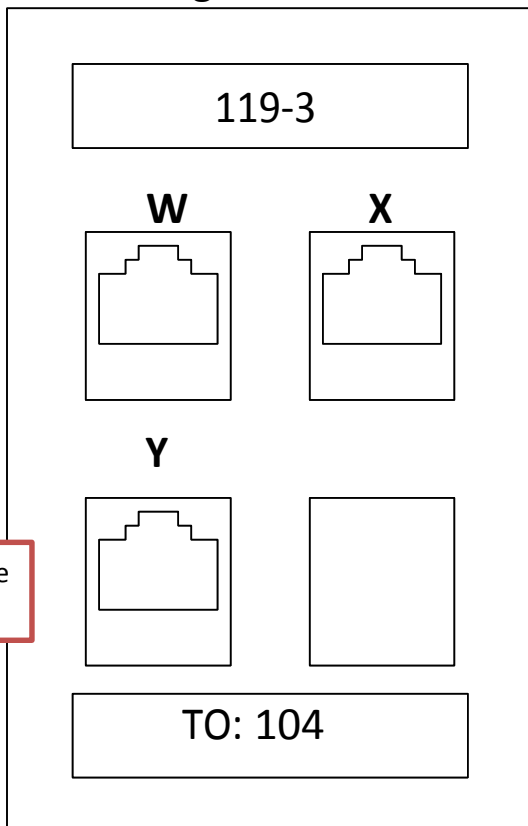


Figure 113

Straight 6 Port

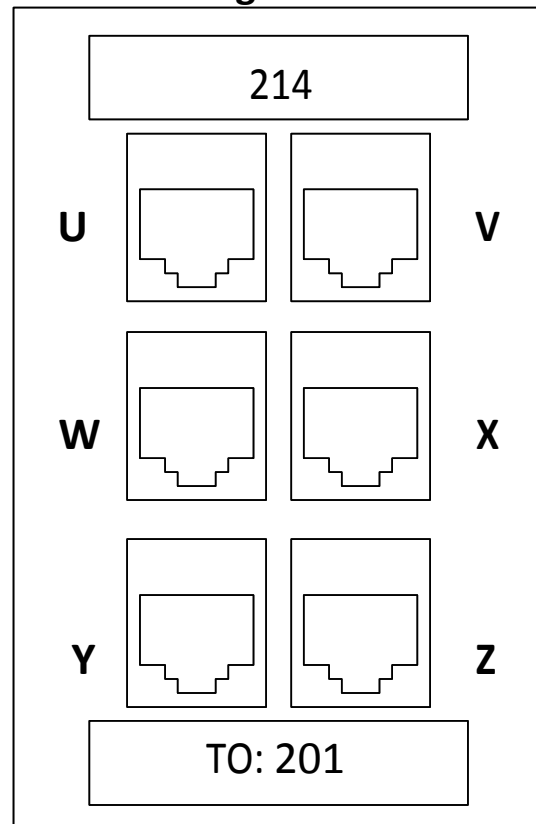
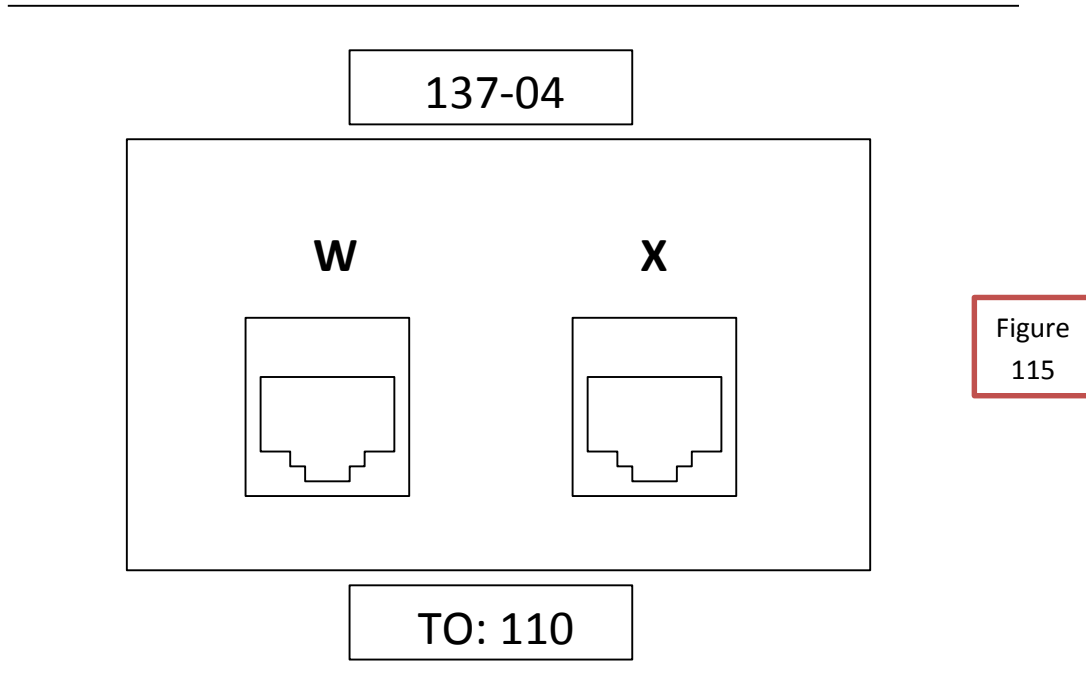
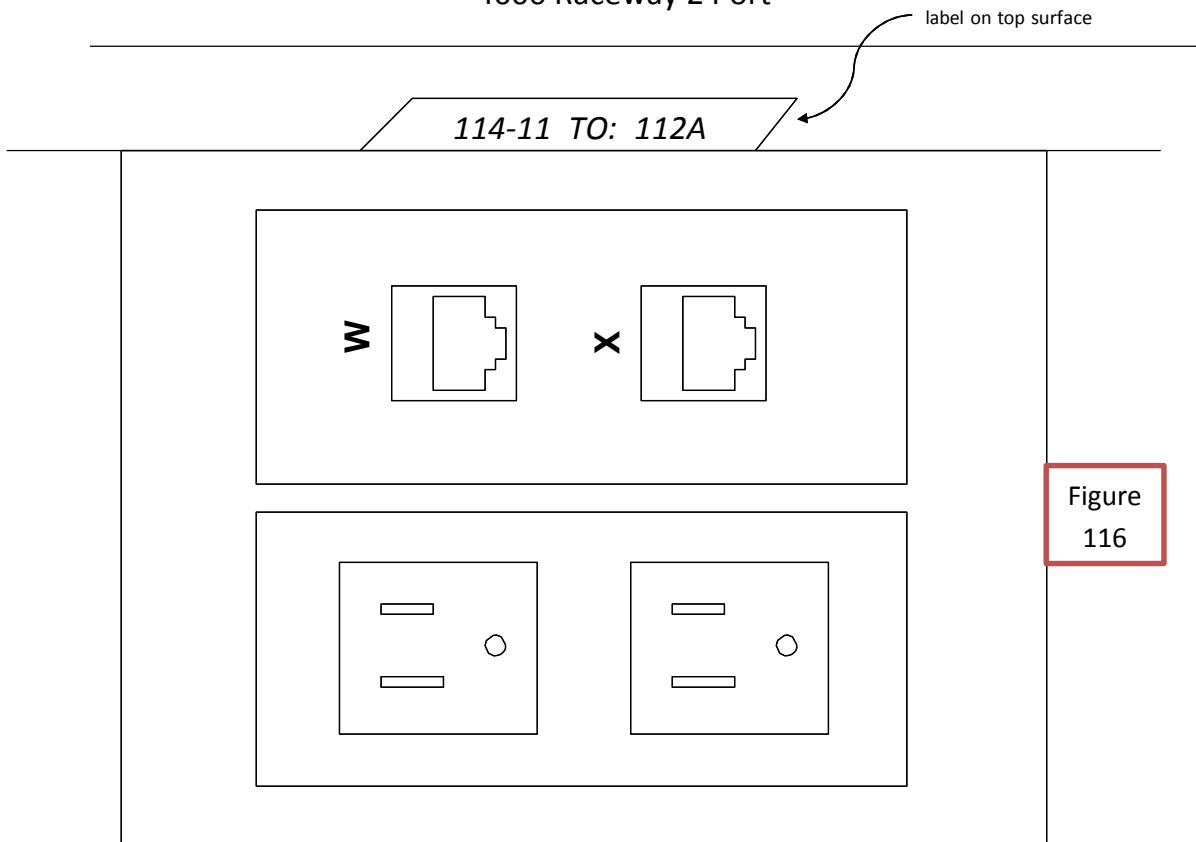


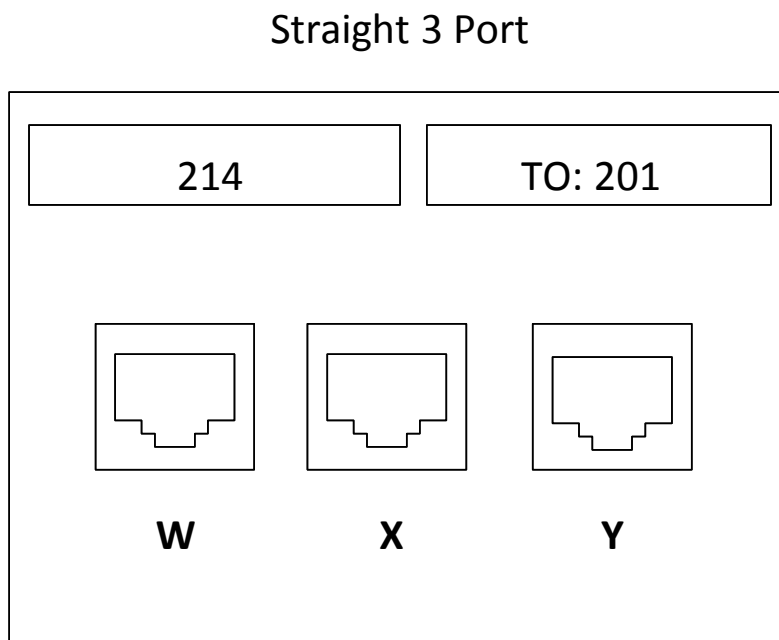
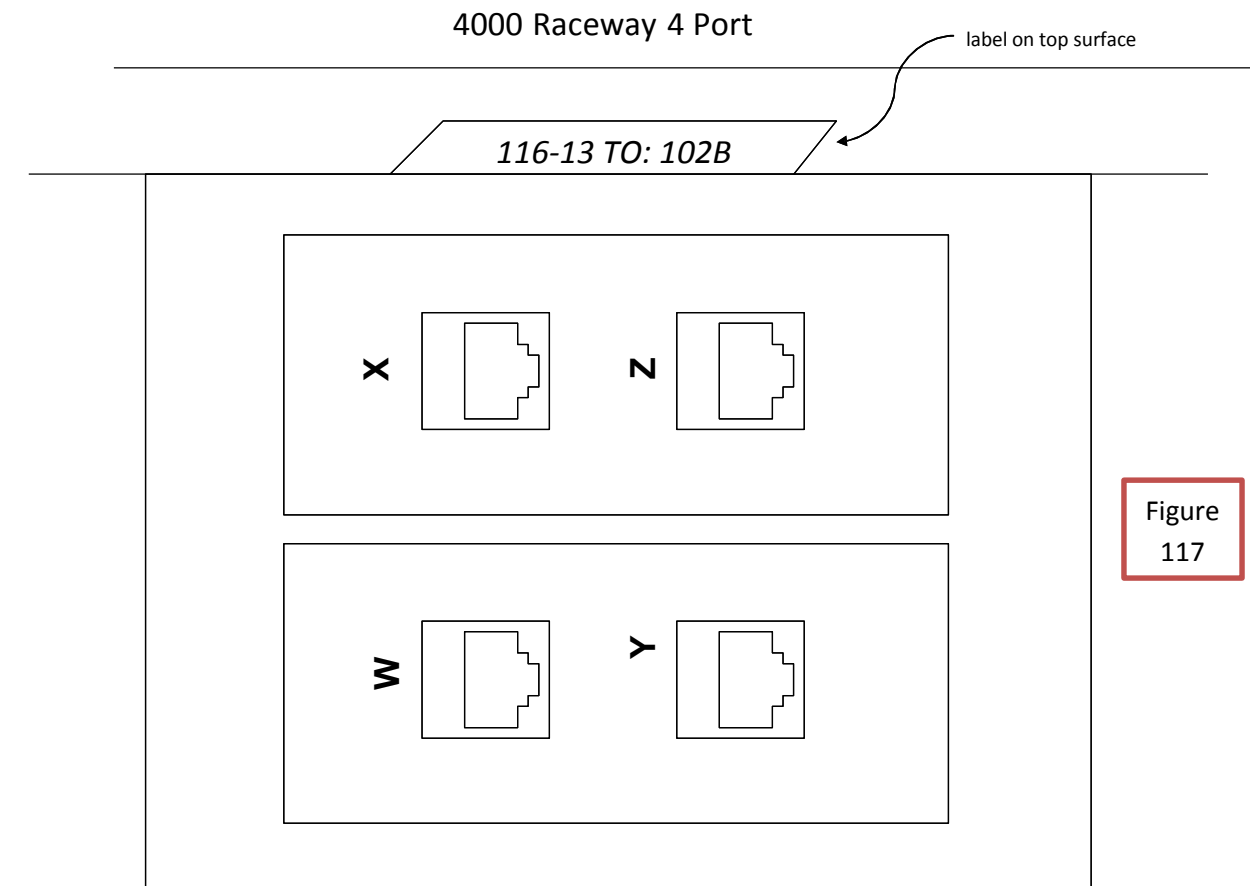
Figure 114

Modular Furniture 2 Port



4000 Raceway 2 Port





CAL POLY I.T.S. ROOM & FACEPLATE NUMBERING SCHEME

FIGURE 1 - ROOM NUMBERING

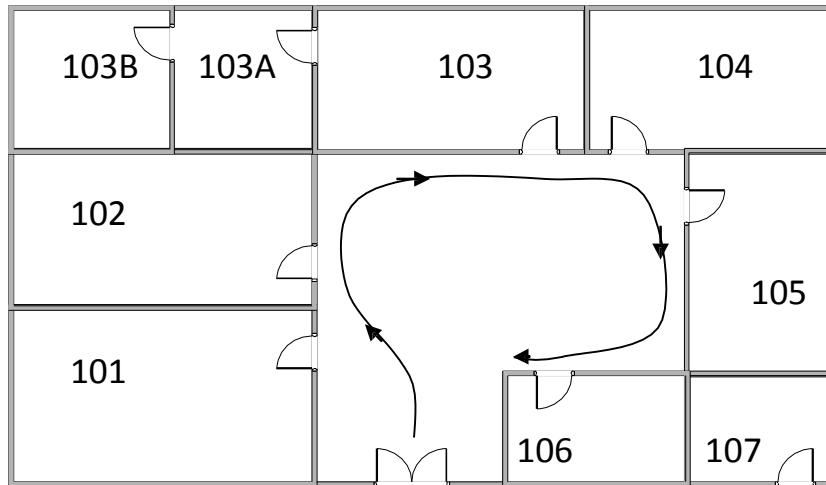


Figure
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LEVEL AND ROOM NUMBERS SHALL BE PROVIDED BY **CAL POLY FACILITIES PLANNING**.

1. ALL ROOMS ON SAME ELEVATION LEVEL SHALL HAVE COMMON PRIMARY DIGIT, I.E., ALL FIRST LEVEL FLOORS ARE "1XX", SECOND LEVEL ARE "2XX", ETC.
2. ROOMS ARE NUMBERED FROM MAIN ENTRANCE IN A CLOCK-WISE PATTERN. ROOMS WHICH ARE ACCESSIBLE ONLY THRU ANOTHER ROOM ARE NUMBERED 'A'-B'-C', ETC. OF THE OUTER ROOM NUMBER (E.G., 103A, 103B, ETC).
3. EXTERIOR ROOMS ARE NUMBERED FOLLOWING HIGHEST NUMBERED INTERIOR ROOM.

FIGURE 2 – FACEPLATE NUMBERING

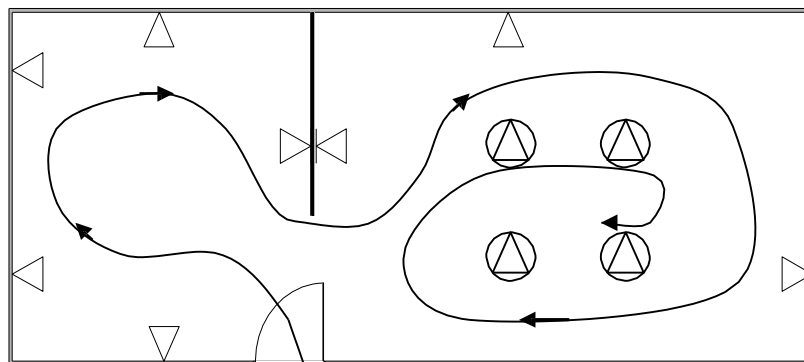


Figure
120

MAIN ROOM ENTRANCE

1. IN ANY GIVEN NUMBERED ROOM, FACEPLATES ON THE WALLS SHALL BE NUMBERED IN A CLOCK-WISE PATTERN BEGINNING FROM THE MAIN ENTRANCE OF THAT ROOM.
2. AFTER NUMBERING ALL FACEPLATES ON WALLS, THEN NUMBER FACEPLATES ON TELE-POLES, FLOOR MONUMENTS OR OTHER "ISLAND" LOCATIONS BY SPIRALLING IN CLOCKWISE TOWARDS THE CENTER OF THE ROOM.
3. REFER TO FACEPLACE LABELING SHEET FOR COMPLETE NUMBER/LABEL SYNTAX.

Cal Poly I.T.S. Telecomm Labeling and Syntax Standards

Location Numbering:

Signal Man Holes:	“ SMH ” followed by a two-digit number, plus a letter, if required.
Signal Pull Holes:	“ SPH ” followed by a two-digit number, plus a letter, if required.
Signal Pull Boxes:	“ SPB ” followed by a two-digit number, plus a letter, if required.
Pedestals:	“ PED ” followed by a two-digit number, plus a letter, if required.
Buildings:	“ B ” followed by a three digit Building number and then a letter where appropriate. See examples below.
Rooms:	“ B ” for Building followed by a three digit Building number with letter if needed, then a “ - ” and a three digit Room number with letter if needed.
Backboards:	If there is only a Backboard on one wall of a Room, no designation is needed. If more than one Backboard exists, then they must be differentiated with a W , E , N or S for W est, E ast, N orth or S outh wall location.

Location Numbering Examples:

- Signal Man Hole / Pull Hole / Pull Box Numbers: **SMH05**, **SPH84**, **SPB103**
- Pedestal Numbers: **PED05**, **PED11**
- Building Numbers: **B001**, **B133**, **B043A**
- Room Numbers: **B001-207**, **B117T-102**, **B047-021K**, **B052-F004**

Room Connection Point Numbering:

Rack Panel Ports:	- RaPbPt xx is a Rack (a), Panel (b) and Port (xx)
Faceplate Jacks:	- aU , aV , aW , aX , aY or aZ are for Jacks on a Faceplate where “a” is the number of the Faceplate counting clockwise around the Room from the Main Entrance.

Note: All of these connection points shall be preceded by a Building and Room designation.

Room Connection Point Numbering Examples:

- Port Numbers: **002-004-R1P2Pt16**, Port 16, in Panel 2, on Rack 1, in Room 004, Building 002.
- Faceplate Jack Numbers: **002-016C-3W**, is the “W” jack on the third Faceplate in Room 016C of Building 002

Conduit Numbering:

Conduits are designated by their Source and Destination locations, Sequence Number of the Conduit in that bank of Conduits and then its Diameter in 1/100ths of an inch.

The order is standardized to have the Lower Numbered Pedestal (PED) first for Pedestal-to-Pedestal Conduits, the Lower Numbered Signal Man Hole (SMH) first for SMH-to-SMH Conduits, the Pedestal first for Pedestal-to-SMH Conduits and the SMH first for SMH-to-Building Conduits.

The order sequences where **PED** = Pedestal, **SMH** = Signal Man Hole, **SPH** = Signal Pull Hole, **SPB** = Signal Pull Box, and **B** = Building are:

PED before **SMH**, **SPH**, **SPB** and **B**

SMH before **SPH**, **SPB** or **B**

SPH before **SPB** of

B SPB before **B**

The syntax is **SMHxx/SMHyy/Dzz-aaa** where xx and yy are the Signal Man Hole numbers, zz is the number of the Conduit in its duct bank and aaa is the diameter of the Conduit in 100ths of an inch.

Conduit Numbering Examples:

- **PED01/SMH13A/D02-400**, is the second Conduit from Pedestal 01 to Signal Man Hole 13A and is 4" in diameter.
- **SMH10/SMH11/D03-300**, is the third Conduit in the Signal Man Hole 10 to Signal Man Hole 11 Duct bank and it is 3" in diameter.
- **SPH02/B108/D03-400**, is the third Conduit from Signal Pull Hole number 02 to Building 108 and it is 4" in diameter.

Cable Numbering:

Cables are identified by their Source and Destination locations. THE ORDER IS IMPORTANT. The Source is the Pedestal for Pedestal-to-Building cables; the Lower Numbered Pedestal or Building for Pedestal-to-Pedestal or Building-to-Building cables; and the Lower Numbered Room for Riser or Station Cables in the same Building.

Outside Plant Cables (OSP):

Twisted Pair Copper Cable Examples:

- **PED04/PED07**, Pedestal 04 to Pedestal 07, Pedestal 04 is the Source and Pedestal 07 is the Destination.
- **PED05/B124-117A**, Pedestal 05 to Building 124-Room 117A, Pedestal 05 is the Source.
- **B002-004/B060-101A**, Cable from Building 002 Room 004 to Building 060 Room 101A.

Fiber Optic Cable Examples:

- **FMB115-000/B124-225A** is a Fiber Optic cable between Building 115 Room 000 and Building 124 Room 225A.
- “**FM**” before any cable designator indicates a **Multimode Fiber Optic** cable; • “**FS**” is a **Single mode Fiber**.
- **FSPED03/PED08-2** where the “-2” indicates a **Second Cable of Multiple** cables between the same two end points.

Coaxial Cable Examples:

- **CPED02/PED03-8** is the **Eighth Coaxial Cable** between Pedestal 02 and Pedestal 03. A “**C**” before any cable designates a **Coaxial cable**.

Note: Buildings have 3-digit Numbers plus a Letter, if required, and Pedestals have 2- digits plus a Letter, if required.

Riser and Station Cable Numbering:

Copper Cable Examples:

- **B010-118/B010-218** is a **Riser Cable** between Rooms 118 and 218 in Building 010.
- **B002-004E/B002-004-S** is a **Cable** between the **East Backboard** in Building 002 Room 004 and the **South Backboard**.
- **B002-004S/B002-004-R1P10-02** is the **second Cable** from the **South Backboard** in Building 002 Room 004 to the **Number 10 Patch Panel on Rack 1** in Building 002 Room 004.
- **B002-004-R1P1Pt16/B002-016C-1W** is a **Station Cable** from **Rack 1, Panel 1, Port 16** in 002-004 to the “**W**” **Jack** on the **first (of two or more) Faceplate** in Building 002 Room 016C.

Fiber Cable Examples:

- **FMB021-123/B021-202** is a **Multimode Fiber Optic Cable** from in Building 021 Room 123 to in Building 021 Room 202.
- **FSB001-501B/B014-103** is a **Single mode Fiber Optic Cable** from Building 001 Room 501B to Building 014 Room 103.

LABELING STANDARDS

BACKBOARDS AND PUNCH BLOCKS

Use this Standard to label Backboards and Punch Blocks in Telecommunications Spaces.

BACKBOARDS: BUILDING-ROOM-WALL

-This Label shall be written with lettering stencils at the top of each installed Backboard.

Example:

-Building 107, Room 100D, South Wall: **BD107-100D-SOUTH**

-Building 053A, Room 100, West Wall **BD053A-100-WEST**

PUNCH BLOCKS:

-Punch Blocks are referenced by their Backboard Number followed by an individual Block Designation Number, with numbering beginning over with each Wall / Backboard in the Telecommunications Space.

-66M-1 Blocks shall be numbered independent of each other from left to right, top to bottom. An exception is when blocks are stacked directly on top of one another. They shall then be numbered in columns.

Example:

Typical Backboard Layout:

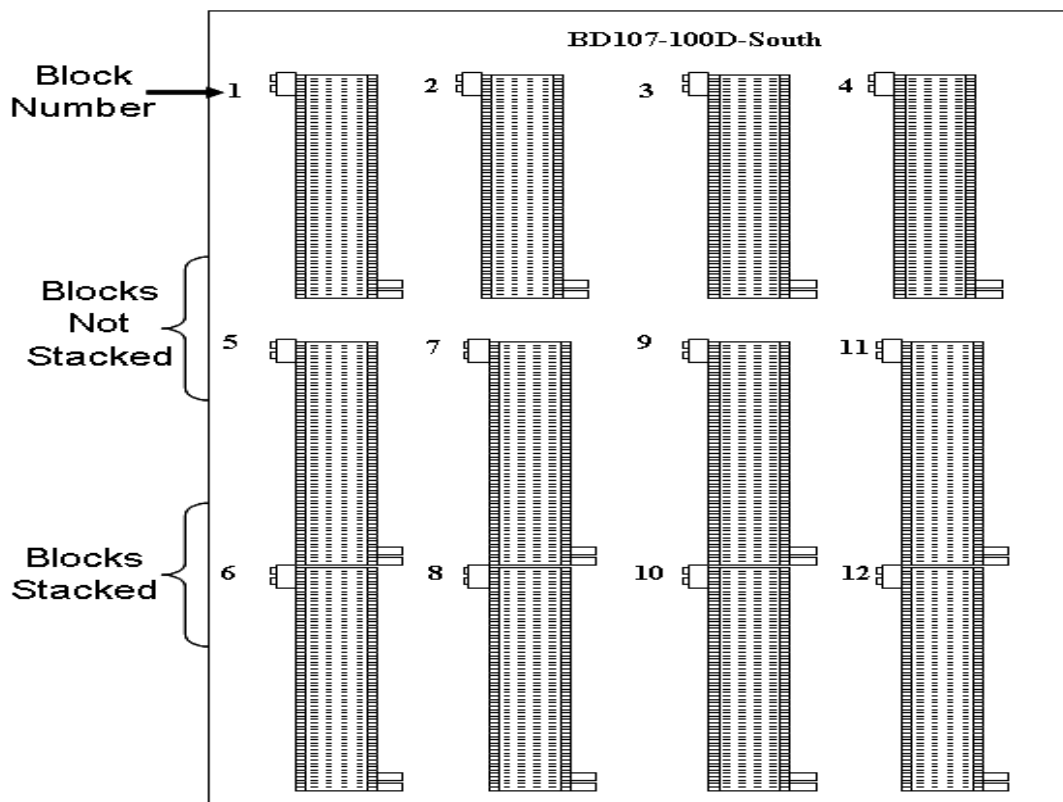


Figure
121

-Example of 66M-1 Block Label Positioning:

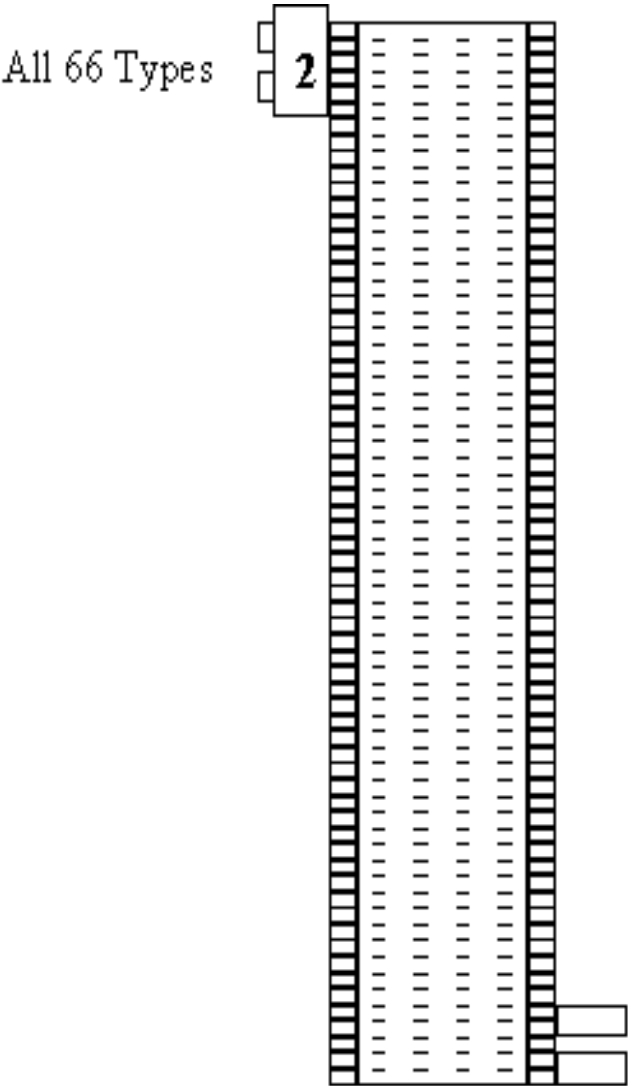


Figure
122

LABELING STANDARDS

RACKS AND PATCH PANELS

This standard should be used for labeling Racks and Patch Panels.

RACKS: *RACK #*

Examples: **RACK 1, RACK 2, RACK 3**, etc.

-Free-standing Racks shall be numbered and labeled from the wall outward, that is, the Rack touching the wall of the Telecommunications Space shall be numbered "RACK 1", and the next Rack out from the wall shall be numbered "RACK2", etc.

-Ideally, the Rack immediately to your left is RACK ONE when viewed from the Racks' front side. With Racks against the right-hand wall of the Telecommunications Space, the numbering will be reversed, but will allow ascending numbers to be used as more Racks are added.

-Wall-Mounted Racks shall be numbered and labeled in ascending numerical order in a clock-wise pattern around the Telecommunications Space, beginning at the main entrance.

-The Label shall be placed at the top of the Rack, if it can be clearly seen. If obscured from view, the label shall be placed on ladder raceway, centered directly above the Rack.

-All labels shall be mechanically produced.

PATCH PANELS: *RACK # PANEL #*

Examples: **R1P1, R1P2, R2P2**, etc.

-Place the Patch Panel Labels centered on the extreme left and right edges on each type of Patch Panel, where space is available. Do not cover individual Port Labeling areas with the Patch Panel number Labels.

-Start at the top of the Rack and label sequentially downward.

-Do not count Horizontal Wire Management, which shall be between each Panel and / or Active Device, plus one each at the extreme top and bottom of all rack equipment.

-Do not count as Panels any Active Devices, such as Hubs, Switches, Routers, UPS's, etc.

-Patch Panels are normally at the top of a Rack while Active Devices are toward the bottom.

-If there are pre-existing Panels (which are already labeled), new Patch Panels are numbered higher since they have been installed later. In other words, do not change the labels of existing Patch Panels just because they are below the new ones that you are installing.

-All labels shall be mechanically produced.

Example: Rack Numbering and Labeling

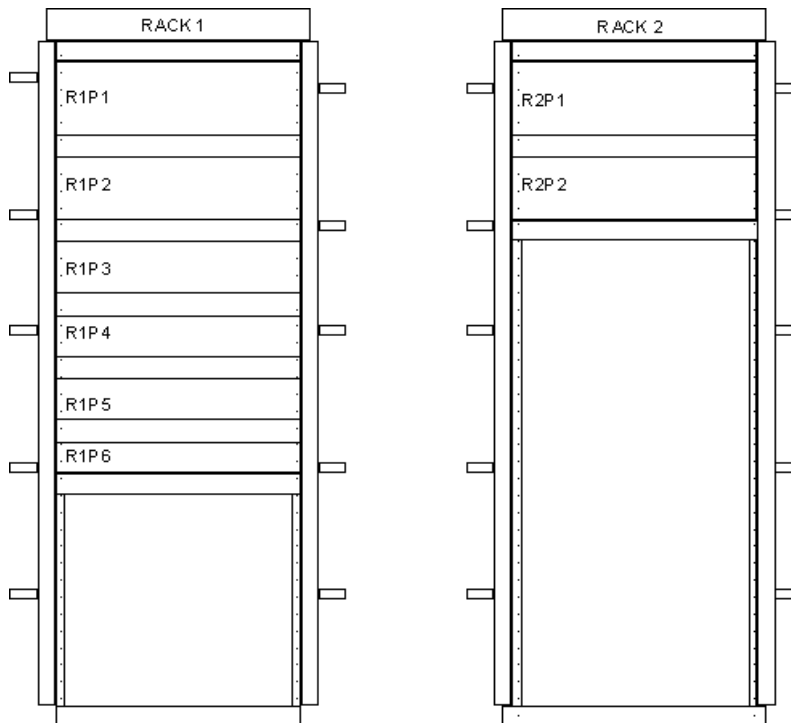


Figure
123

PATCH PANEL PORT LABELS:

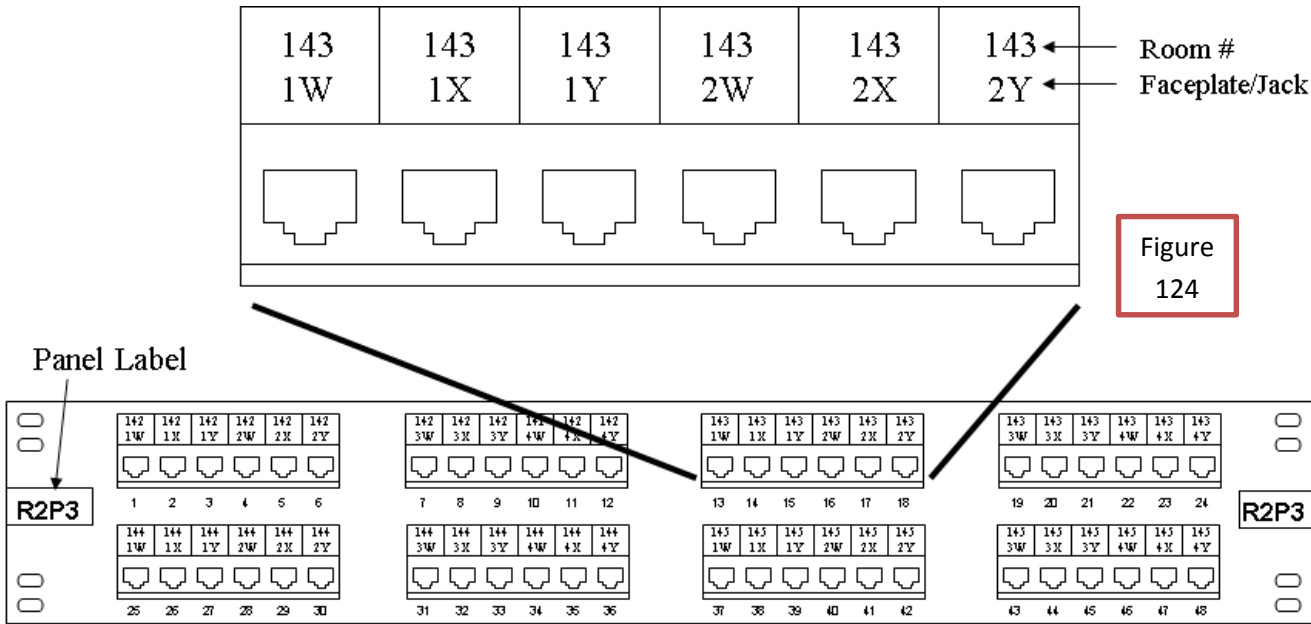
Templates for Patch Panel Labels are available at: 'S:\ITS\CCS\TECH\Documentation\AllCircuit Documentation Sheets\All Templates for Documentation'.

-Each individual Port Label shall consist of the Room and Faceplate / Jack Number that Port connects to. Style of Labels shall match Patch Panel and will vary by manufacturer.

-Individual Ports on a Patch Panel are named: **Rack#Panel#Port##: R1P1Pt01, R2P3Pt18.**

-All Patch Panel Labels shall be mechanically produced.

Example:



CAMPUS STANDARD POSITIONS OF ITEMS IN TELECOMM RACK:

Generic Rack Layout with 1 Rack

FOR FIBER OPTIC TERMINATION BOXES:

PLACE CAMPUS FIBER BACKBONE CABLE FIRST. REMAINING CABLES FOLLOW IN **ASCENDING NUMERICAL ORDER** BY BUILDING/ROOM NUMBER.

FOR ALL FIBER CABLES WITH A COMMON END POINT, PLACE MM FIRST, SM SECOND.

ADD OR LEAVE 25% FREE FOR FUTURE GROWTH.

IDEAL COMPONENT LAYOUT IN SINGLE RACKS SHALL HAVE:

1. ONE RACK UNIT OPEN AT THE TOP (TO CLEAR THE "J" HOOKS).
2. FIBER OPTIC TERMINATIONS (BOX, PANELS, SHELVES, ETC.).
3. COPPER PATCH PANELS, EACH SEPARATED BY A DOUBLE WIRE-MANAGEMENT UNIT.
4. ACTIVE DEVICES (HUBS, SWITCHES, ROUTERS, ETC.).
5. RACK SHELVES SHOULD BE ATTACHED AT THE LOWEST RACK UNIT FOR ITEMS SUCH AS U.P.S.'s, P.C.'s, SERVERS, ETC.

VERTICAL WIRE MANAGEMENT, FRONT & BACK STYLE BOTH SIDES OF EACH RACK. TYP.

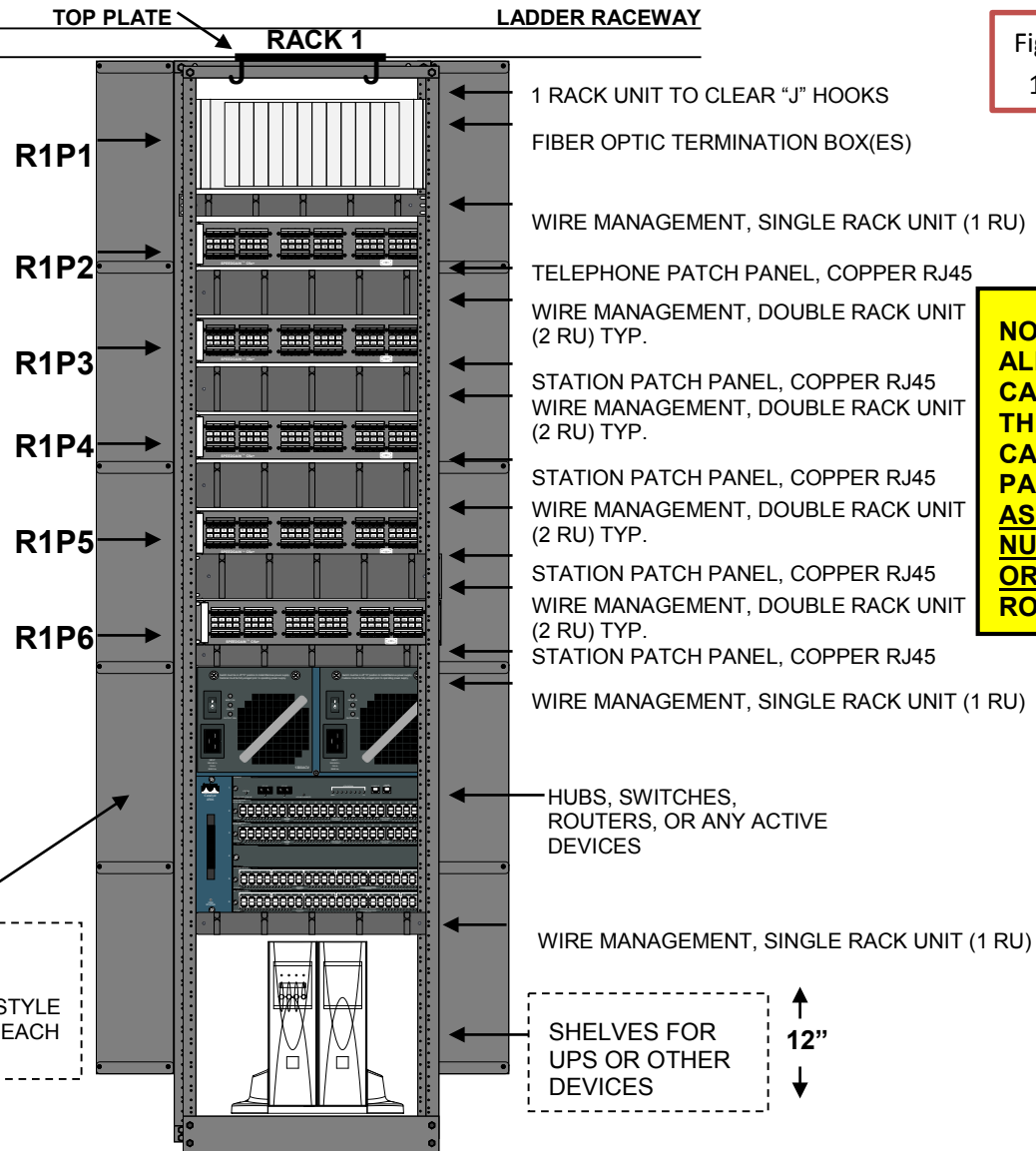


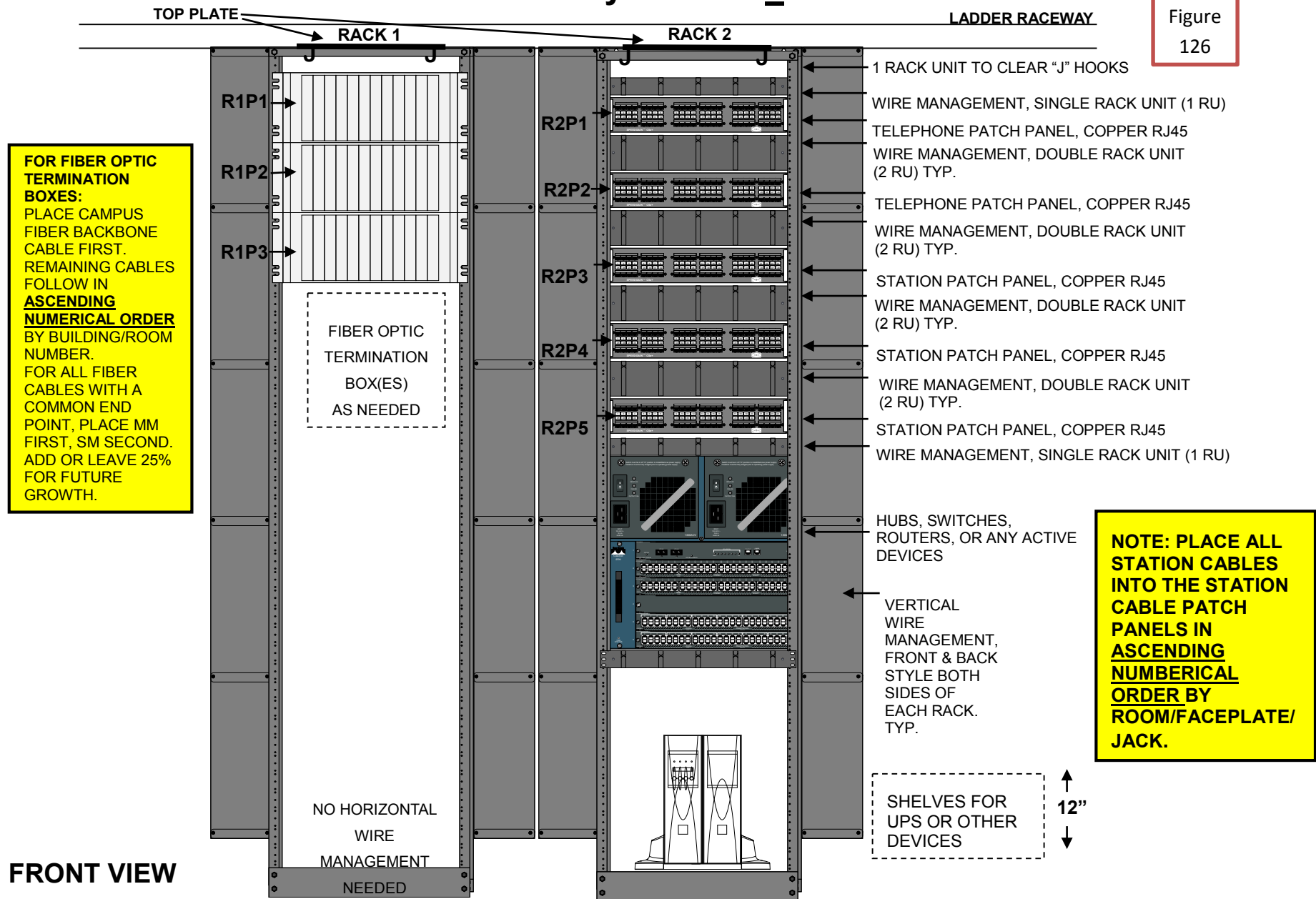
Figure
125

NOTE: PLACE ALL STATION CABLES INTO THE STATION CABLE PATCH PANELS IN ASCENDING NUMERICAL ORDER BY ROOM.

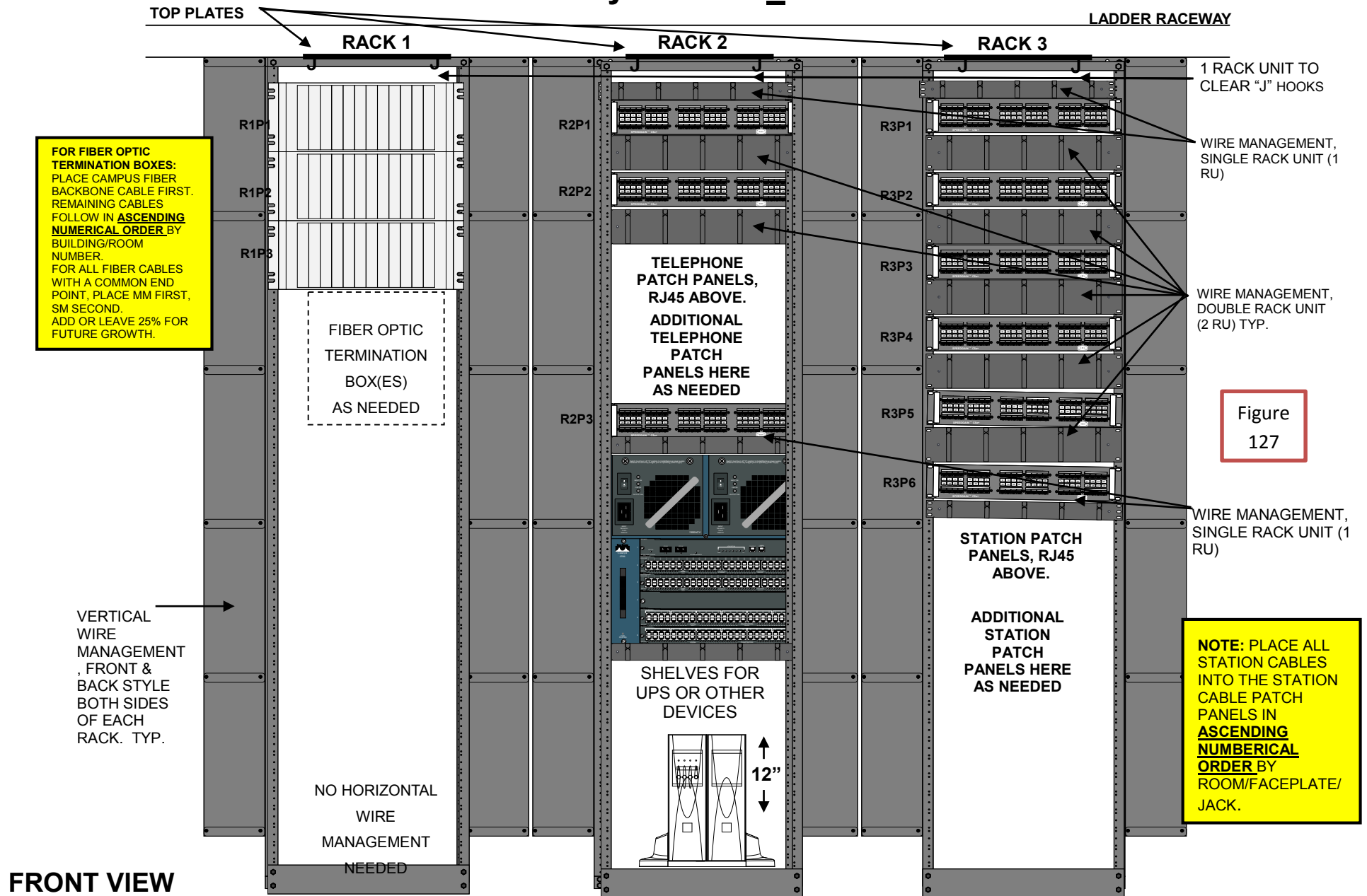
FRONT VIEW

CAMPUS STANDARD POSITIONS OF ITEMS IN TELECOMM RACK: Generic Rack Layout with 2 Racks

Figure
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CAMPUS STANDARD POSITIONS OF ITEMS IN TELECOMM RACK: Generic Rack Layout with 3 Racks



CAMPUS STANDARD POSITIONS OF ITEMS IN TELECOMM RACK

Generic Rack Layout with 4 Racks

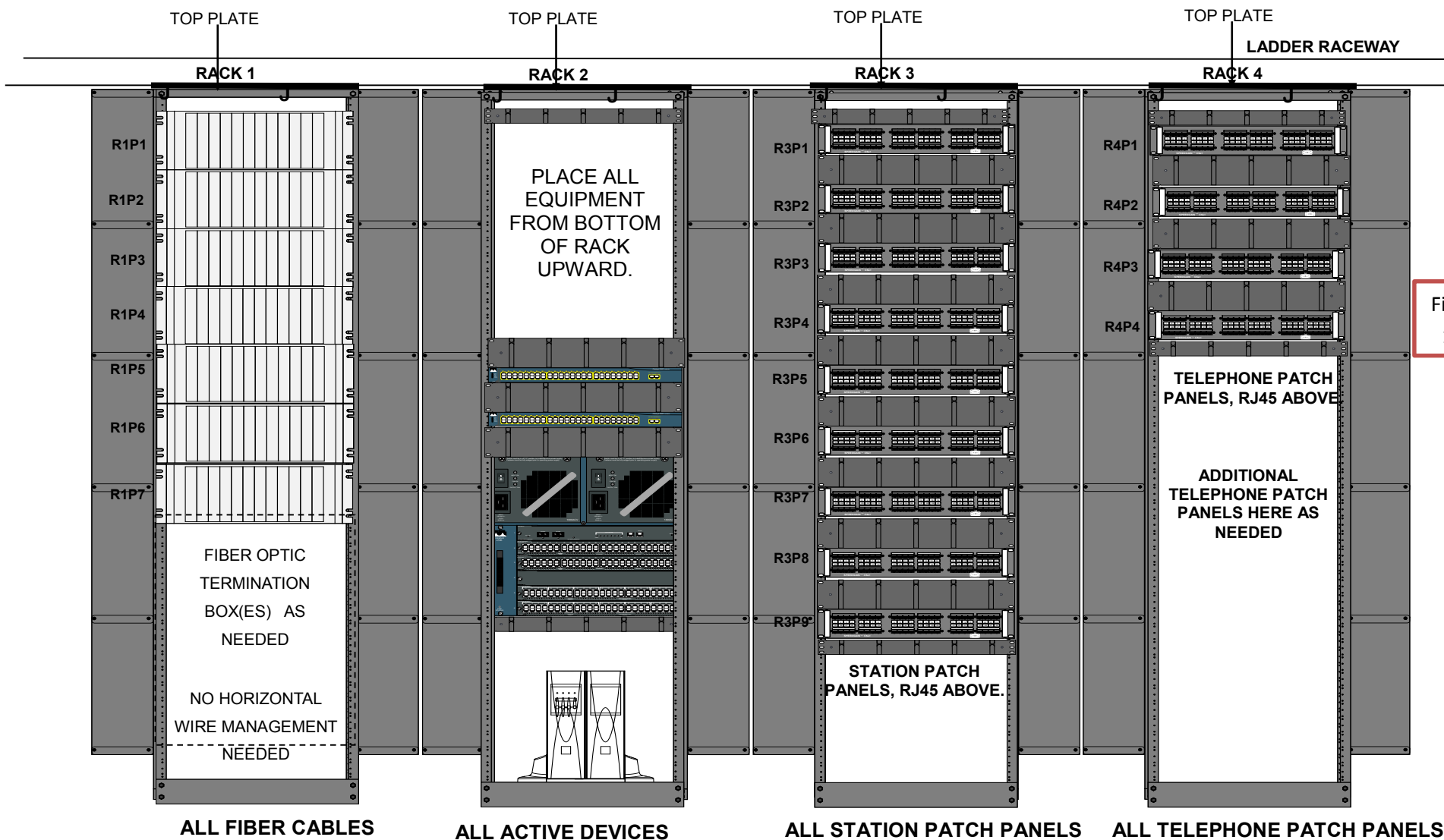


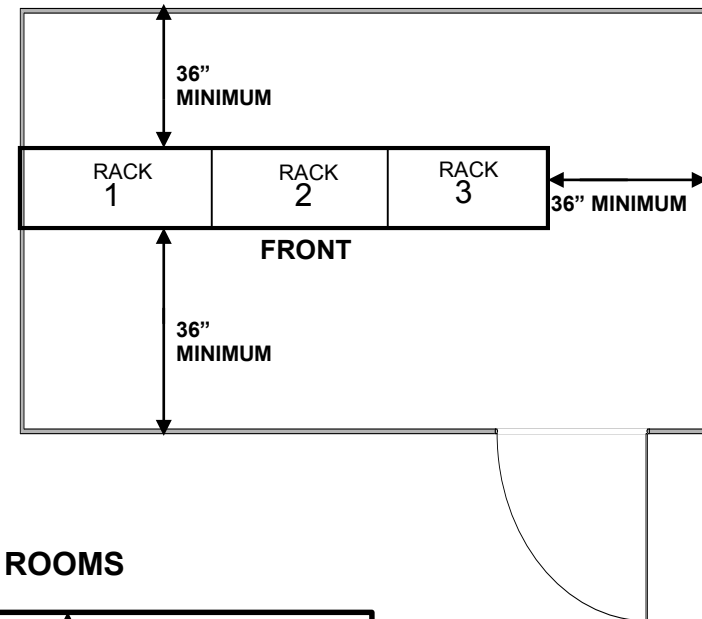
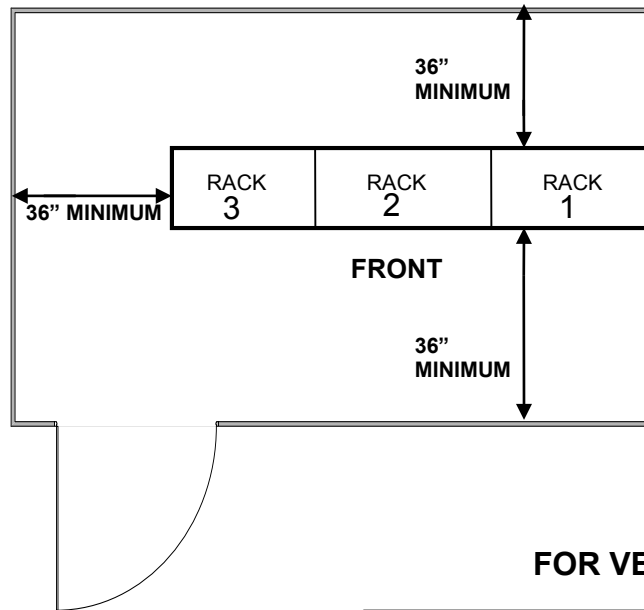
Figure
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FRONT VIEW

RACK NUMBERING & POSITION IN TELECOMM ROOMS

NUMBER RACKS FROM WALL OUTWARD, AS INDICATED.

ALIGN RACKS WITH LONG AXIS OF RECTANGULAR ROOMS.



FOR VERY NARROW ROOMS

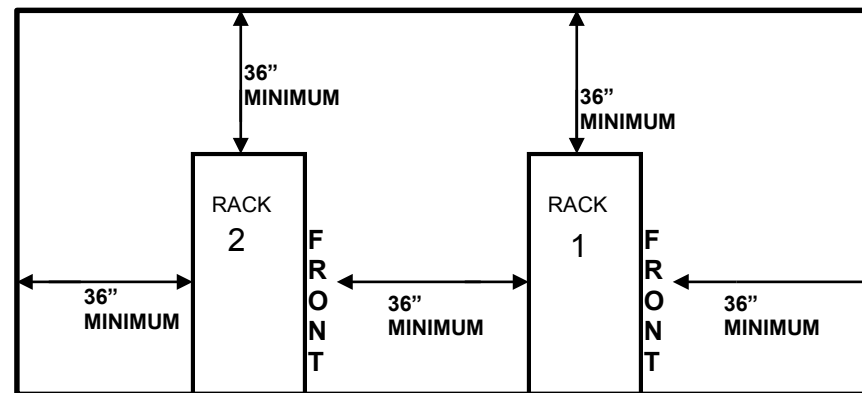
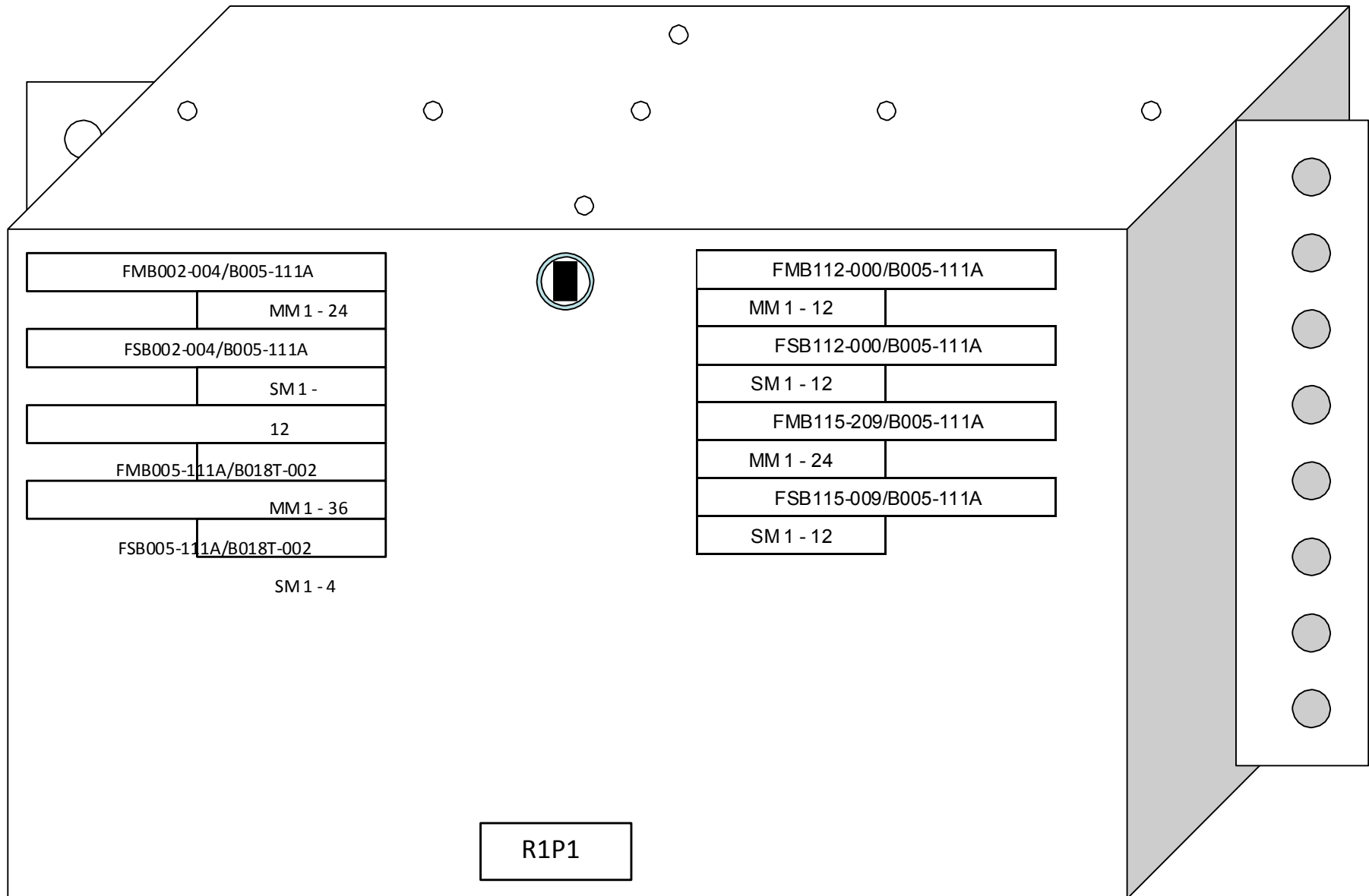


Figure
129

NOTE: THE "FRONT" AND "REAR" 36" DISTANCE RELATES TO THAT OF THE FURTHEST EXTENDING EQUIPMENT

FIGURE 130. FIBER OPTIC CABLE TERMINATION SHELF-EXTERNAL LABELING

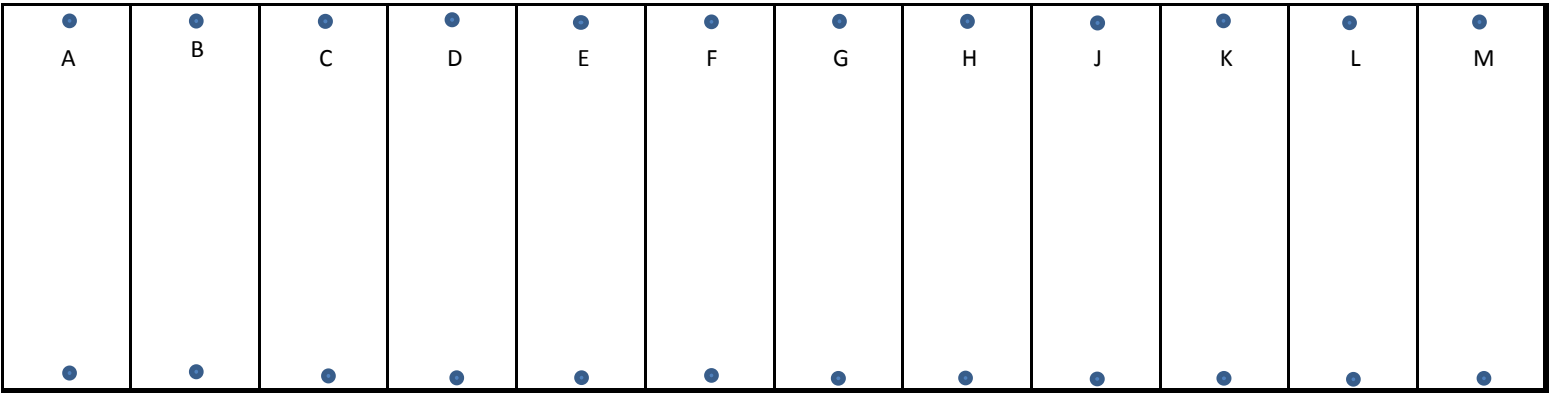
Figure
130



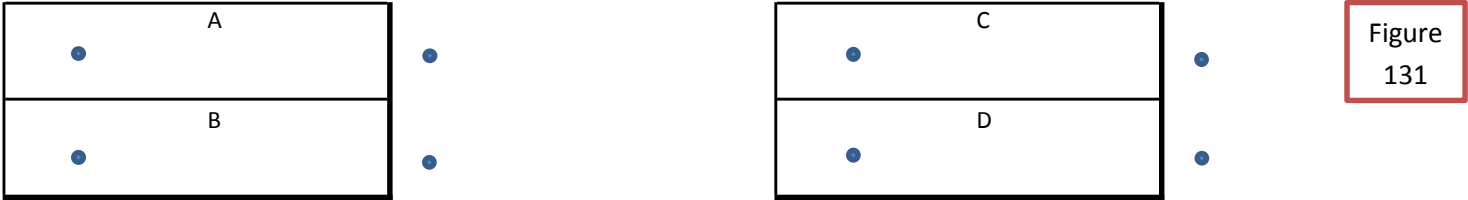
FIBER COLOR - ORDER FOR PATCH PANEL BULKHEAD VARIANTS

BULKHEAD POSITION NOMENCLATURE

ITEM 1 - VERTICAL - VEIWED FROM FRONT



ITEM 2 - HORIZONTAL - VEIWED FROM FRONT



ITEM 3 - HORIZONTAL - VEIWED FROM FRONT

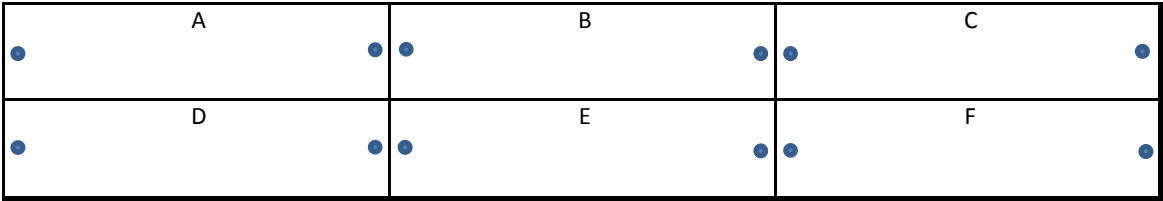




Figure
132

Battery back-up configurations vary. Consult with the Cal Poly ITS Telecomm group for the proper configuration for each project.



Figure
133

Power wiring for equipment installed/mounted in two post racks in every TR/ER shall be mounted to the ladder racking above and to the rear of the racks and done so in such a way as to not block the use of any rack unit locations.

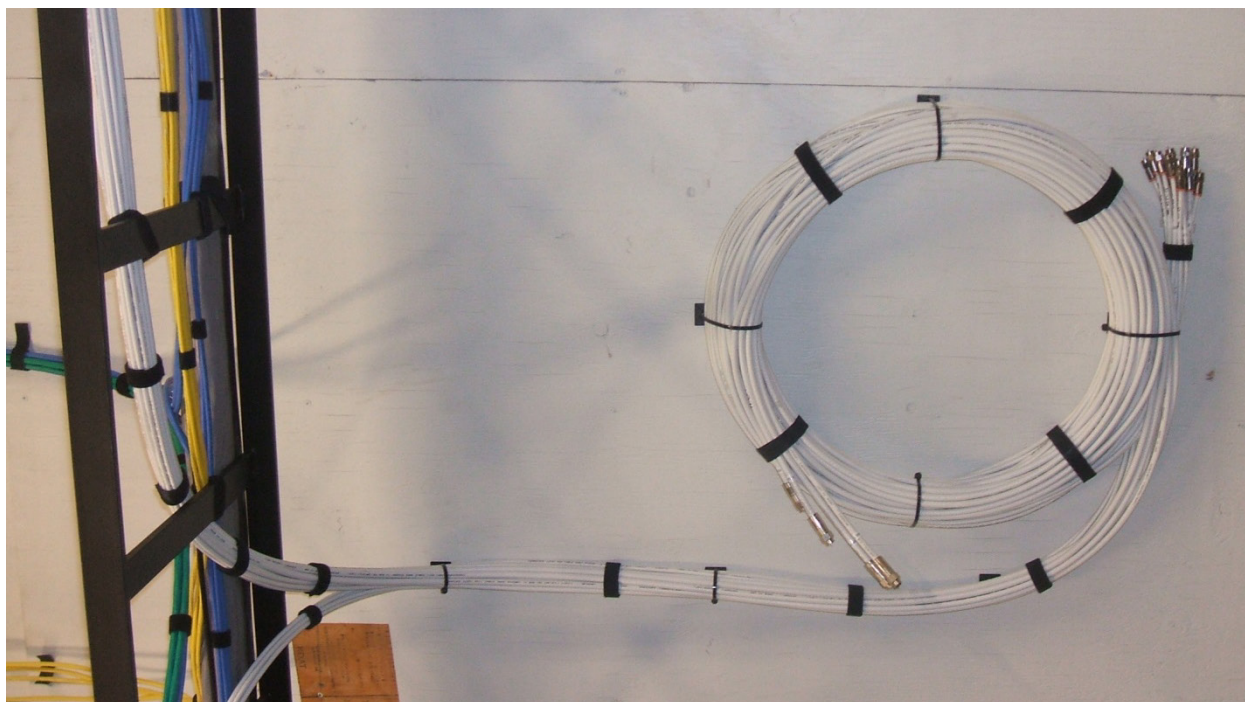


Figure
134

Show is the proper method for providing CATV coax cable for future use.

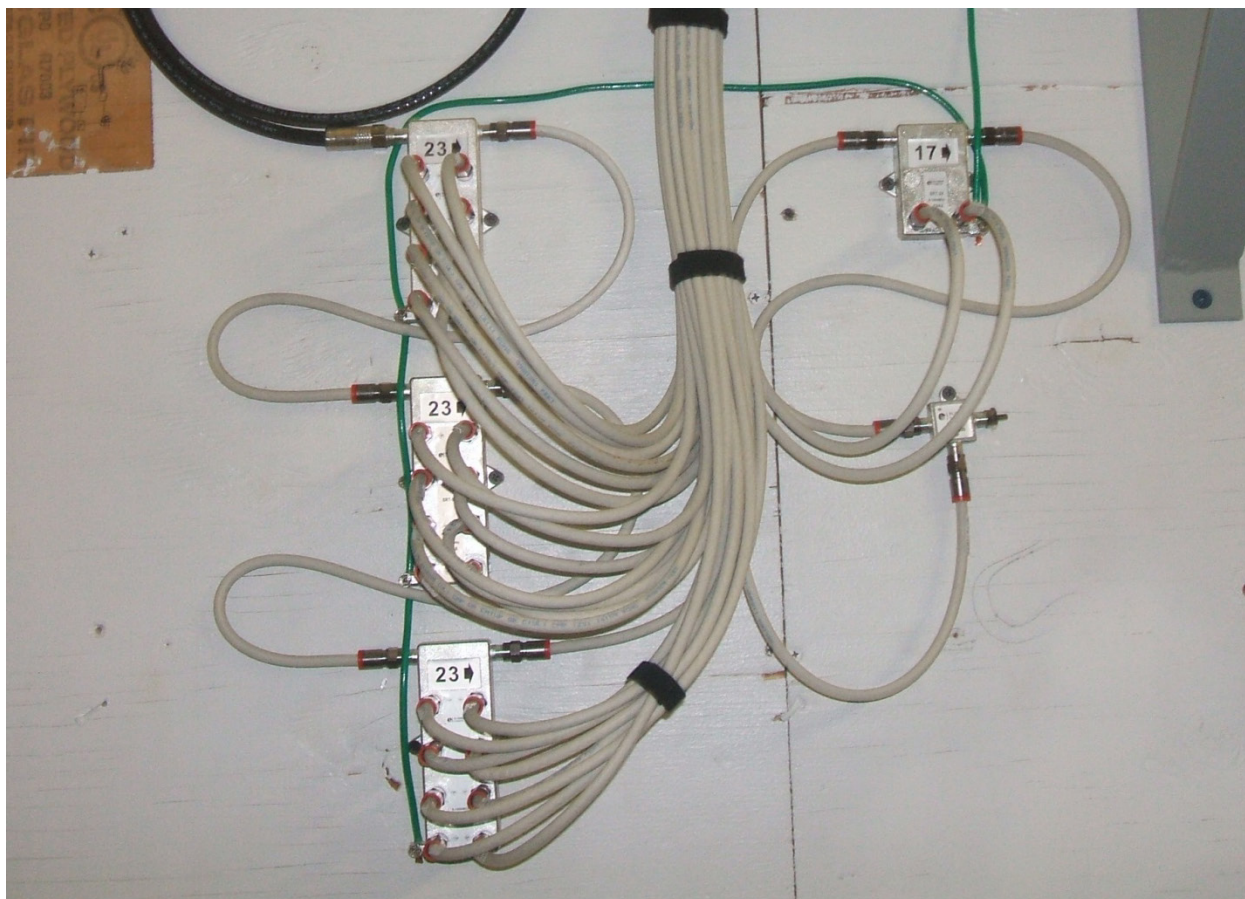


Figure
135

Dress CATV connections neatly on the TR/ER rear wall (opposite the wall containing the telecomm66 “block field”). Be sure to leave an adequate slack loop in every TR/ER.



Figure
136

Seal conduits (including bushings) containing cables properly. Conduits must be re-enterable and where required, sealed using a UL approved methodology to maintain required fire ratings. Non-fire rated sealing shown above.



Figure
137

Seal empty conduits using compression plugs as shown.



Figure
138

Conduits penetrating fire rated walls shall be sealed using materials approved by the State Fire Marshal. Where possible use removable pillows for sealing conduits to make them easier to re-enter if necessary in the future.

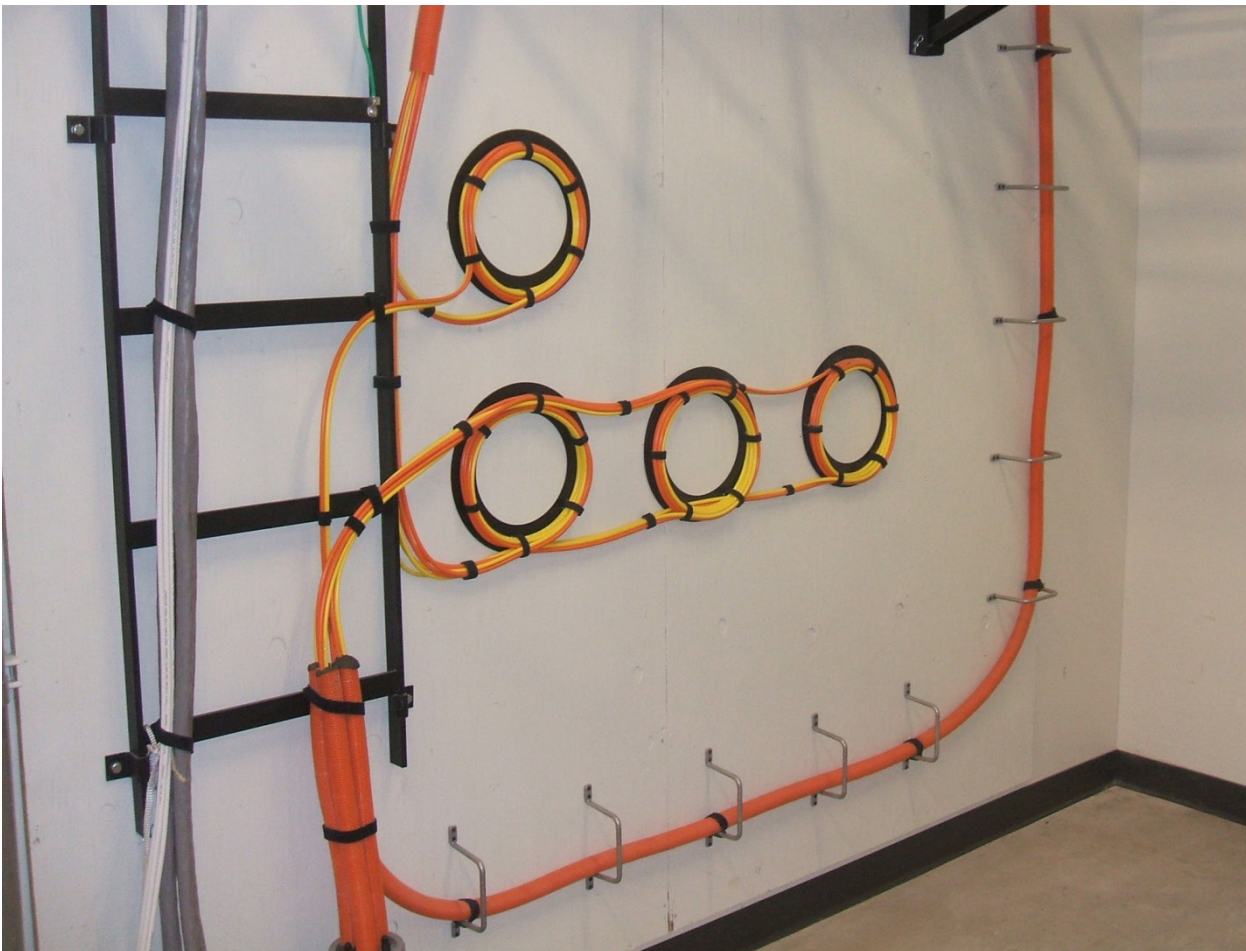


Figure
139

Shown is the proper method for providing fiber optic cable slack loop in a TR/ER.

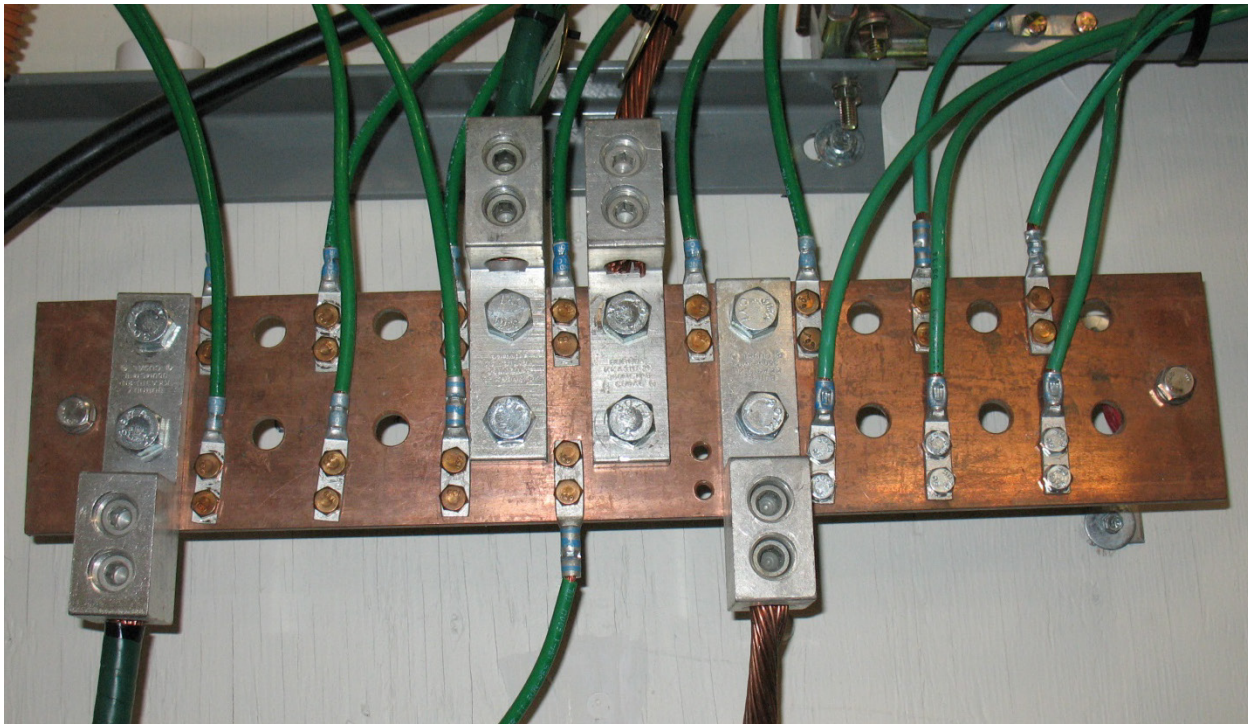


Figure
140

Proper UL Listed grounding busbar. Note that all connections require the use of two-hole mounting lugs. Also note that the large lugs are incorrect and should use crimped-on type lugs to be correct. Use only permanently attached lugs for ground connections.

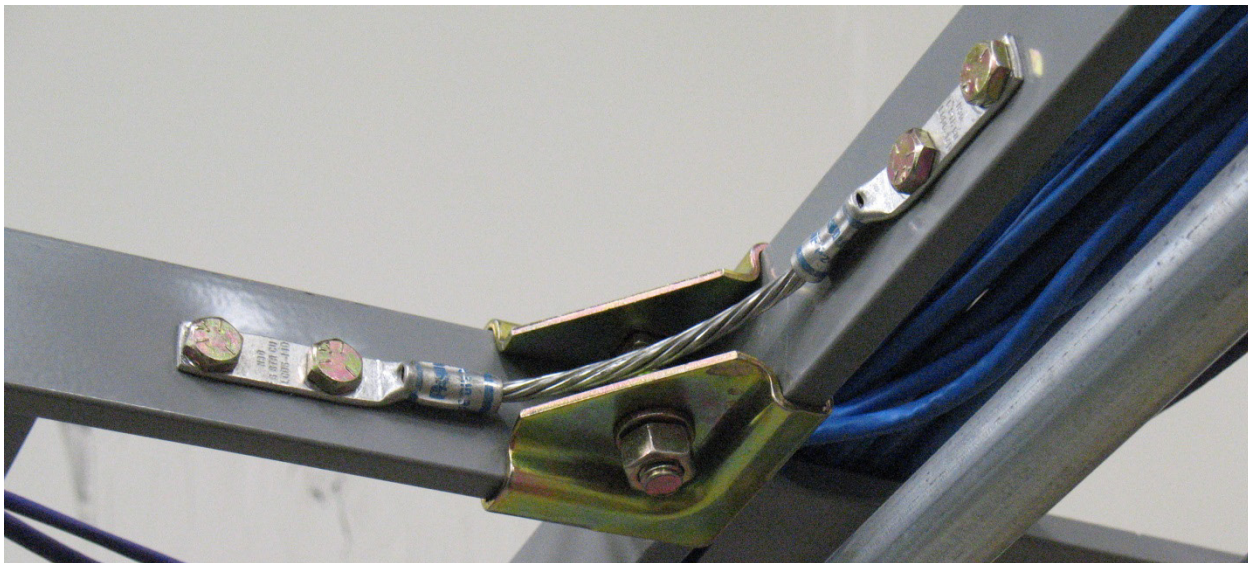


Figure
141

All ladder racking, cable tray and equipment racks must be properly bonded and grounded. Use star washers on painted surfaces as well as scraping away any paint to assure a good ground connection.



Figure
142

Protectors shall be used any time phone lines leave the drip line of the building and again when they enter their new destination. Protectors must be grounded and bonded using a minimum #6 copper ground wire.



Figure
143

Telephone punch-down, 66 blocks shall be mounted to the back board on the (front) wall across from the front of the equipment racks. The “block-field” wall shall also be across from the (rear) wall containing the CATV connections.



Figure
144

Equipment racks shall be mounted side-by-side in a straight line, starting a minimum of 2" from the nearest wall; be plumb, level and square; and allow a minimum of 36" of clear working space behind and in front all mounted equipment. Racks shall be individually grounded to the grounding buss bar found in each TR/ER.



Figure
145

Equipment racks shall be made of bare aluminum and include 12" of double-sided vertical wire management between racks and 6" double-sided vertical wire management on each end. Covers for the wire management shall also be provided. Ladder rack (18") shall be installed above all equipment racks.



Figure
146

Back board in the EFs/TRs/ERs shall leave the fire rating stamp visible at all times. The balance of the back board shall be painted with 2 coats of a white paint.

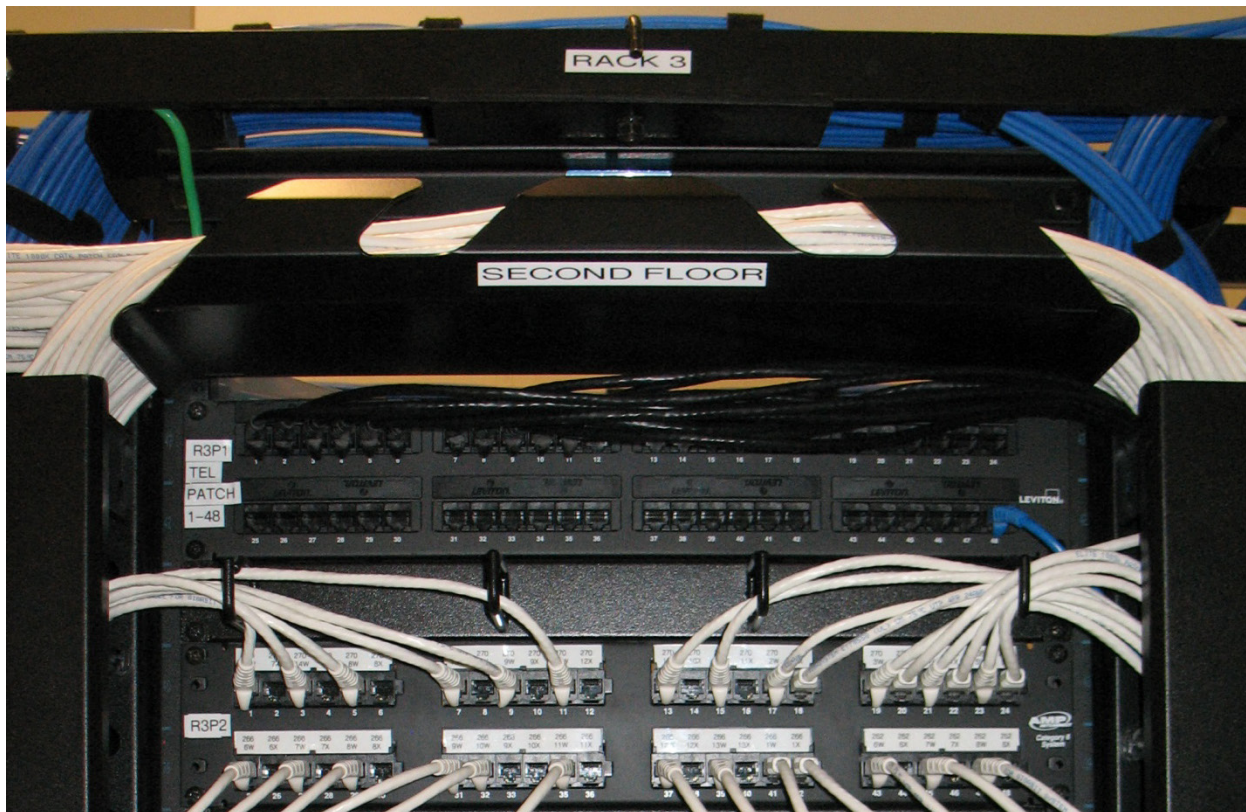


Figure
147

Rack cabinets and patch panels shall be labeled using permanent white labels with black lettering.

Telecomm Room Design Considerations and Requirements

This is the area of the TR/ER that is best suited to house the required telephone protectors. This would be the location where incoming OSP (Outside Plant Cable) is terminated before becoming inside, riser UTP (Unshielded Twisted Pair cable) used for voice/telephone connectivity within a structure. The protector provides protection against short circuits, voltage spikes and other anomalies that could cause damage to the sensitive electronics that handle today's heavy telephone traffic.

This is the area of the TR/ER that is best suited to house the required "66 block", telephone punch block field. Wiring, coming from the protectors mounted nearby, must be punched down on blocks to mate with wiring that comes from the telecomm patch panels in the equipment racks. Once the phone signal is available at a jack on the patch panel, it can, via a short patch cable, be connected to the structured (CAT 6) patch panel cabling that finally routes the signal to the modular jack on the faceplate in the room where the phone is located.

This is the area of the TR/ER that is best suited to locate the required Telecomm Grounding Busbar (TGB). The busbar assembly is a UL listed assembly to which all telecom, data and CATV equipment and pathways must be connected and which is connected to the Main Telecomm Grounding Busbar (TMGB) in the ER. Bonding together all equipment and pathways creates an entire system at the same ground potential, thereby reducing the possibility of "noise" or ground potential differences that can cause a system to fail or work incorrectly.

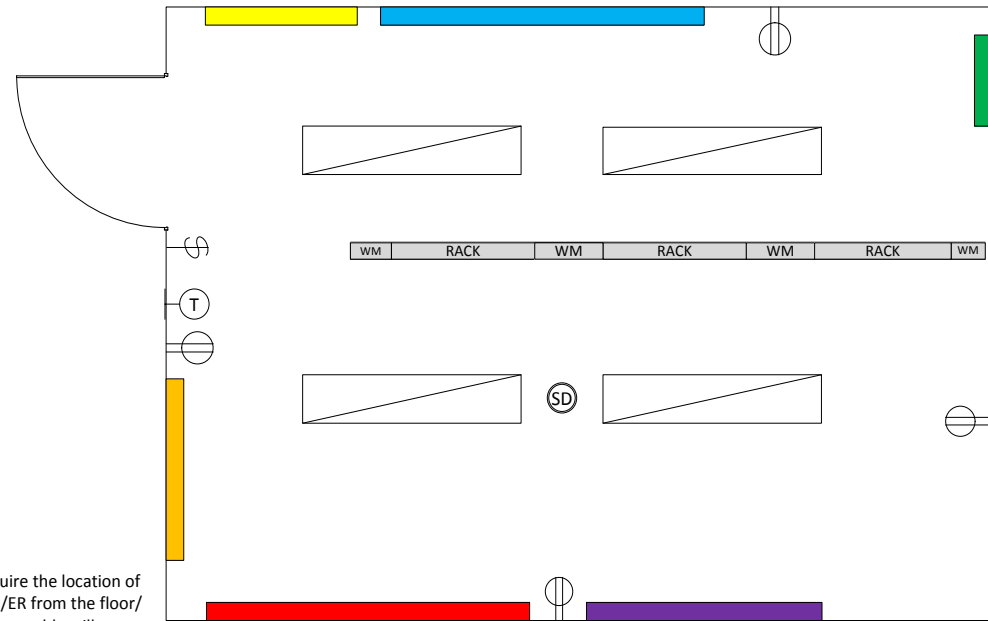
This is the area of the TR/ER that is best suited to house CATV equipment and components if required. CATV is provided to a project or building on fiber optic cable. If multiple buildings are involved, the CATV signal will arrive from the "head end" on fiber to an optical splitter where it will be split and forwarded on by fiber to a "fiber node" in the building it will serve. The node converts the information from fiber optic to copper coax cable. In the ER/TR the building's copper based signal is then run through a series of splitters and taps to finally provide the signal that arrives at the "F" connector in the room where TV viewing is desired.

NOTES:

1. Be sure that the equipment complement for the room is defined so that 3' of clearance can be provided in front and behind the farthest protruding rack mounted equipment.
2. The door must be a minimum of 3' wide and open into a public space (not a living, storage, office or classroom space).
3. Leave a minimum 2" gap between the wall and the first rack (for future cabling).
4. One light fixture (minimum) shall be battery/emergency power backed-up.
5. Power distribution for rack-mounted equipment shall be attached to the ladder racking above the rack (see Figure 133 in Appendix B)

This is the area of the EF/TR/ER that we require the location of conduits (with bushings) entering the EF/TR/ER from the floor/slab. OSP/UTP (CAT 3 twisted-pair) telephone cable will typically be connected/spliced, within a splice case, to ARMM, plenum rated, wiring for distribution throughout the building. The conduits shall enter the room within 3" of the associated wall. As the OSP cabling could be terminated on a protector or spliced and run within the building to the ER, submit shop drawings to the ITS Telecomm group in advance showing the proposed method of CAT 3 cable distribution for approval.

This is the area of the TR/ER that is best suited to house the required electrical power distribution panel that controls all circuits housed within the EF/TR/ER. Never install the panel in the middle of a wall as it will waste too much space. Breakers for the AC unit that cools the room must also be housed in this panel. A close association of the power panel to the entry door allows for easier resetting of a breaker by an electrician, with or without room lighting available.



The Design, Construction and Approval Process

When a construction or remodeling project is deemed necessary and the planning process begins, the ITS Telecomm group should be contacted and your communications needs discussed. The CSU System, through the Cal Poly ITS Telecomm group, has the desire and obligation to help you plan for your current and future needs as they pertain to Telecommunications Systems and Services. Technology is constantly changing and the CSU System is mandating changes and equipment refreshes at times and in ways you may not be aware.

The cost of construction is of course important, but the cost of maintaining and servicing is also substantial and on-going. We will work with you to develop a system design and equipment suggestions in the areas of Telephone, Data, CATV, WIFI and Radio Communications Systems that will be cost-effective over the long-term. The earlier we are involved the better and more comprehensive the result will be.

To offer the greatest level of support we can, the Cal Poly ITS Telecomm group has developed a **Telecommunications Standard Document (TCD)** for the Cal Poly Campus. It is the most definitive document describing how to do all things Telecomm on campus. The TCD is available on-line 24/7 at www.calpolyitstelecomm.edu. This 4 page "Cheat Sheet", the Labeling, Design and Syntax Standards, the Campus Building Numbering and Naming Listing and more are available for download or printing at your convenience. The documents and data provided will be used by us to evaluate if the work done meets the developed standards. If you have any questions you can contact **Tech Services** at (805) **756-7315**. If we do not hear from you we will assume that you fully understand the expectations and requirements of the CSU System and the Cal Poly ITS Telecomm group.

We look forward to working with you to create the greatest value for all parties.

Figure
148



Figure
149

Provide adequate power in every TR/ER. The power panel shall control all circuits within the TR/ER only. ERs shall have a 225 amp, 208 volt AC, 3 phase power panel and TRs shall have a 125 amp, 208 volt, 3-phase power panel.



Figure
150

Wireless Access Points can usually be mounted indoors vertically on a wall or horizontally on the ceiling. All WAPs are to be labeled in a manner similar to the WAP shown above.



Figure
151

WAPS can be mounted outdoors in a protective, weather-tight enclosure attached to a building.



Figure
152

Code Blue/ Talk-A-Phone phones are typically located in bollards throughout the Campus.



Figure
153

Code Blue/ Talk-A-Phone phones also come in a wall mount version where circumstances require wall mounting.



Figure
154

Building/Project Main Equipment Rooms shall be supported by generator/automatic transfer power backup, systems if available.



Figure
155

Above is a typical “Local” backup generator system using natural gas as a clean fuel source. The generator feeds power to the automatic transfer switch thereby providing emergency power to critical equipment.



Figure
156

Shown is a steel Communications Signal Pull Hole cover provided for the Poly Canyon Village Student Housing Project. This is a sample of the type of manhole cover required for Signal Pull Holes (manholes).



Figure
157

Above is a blowup of the nomenclature style to be used when providing Signal Pull Hole (manholes) covers.

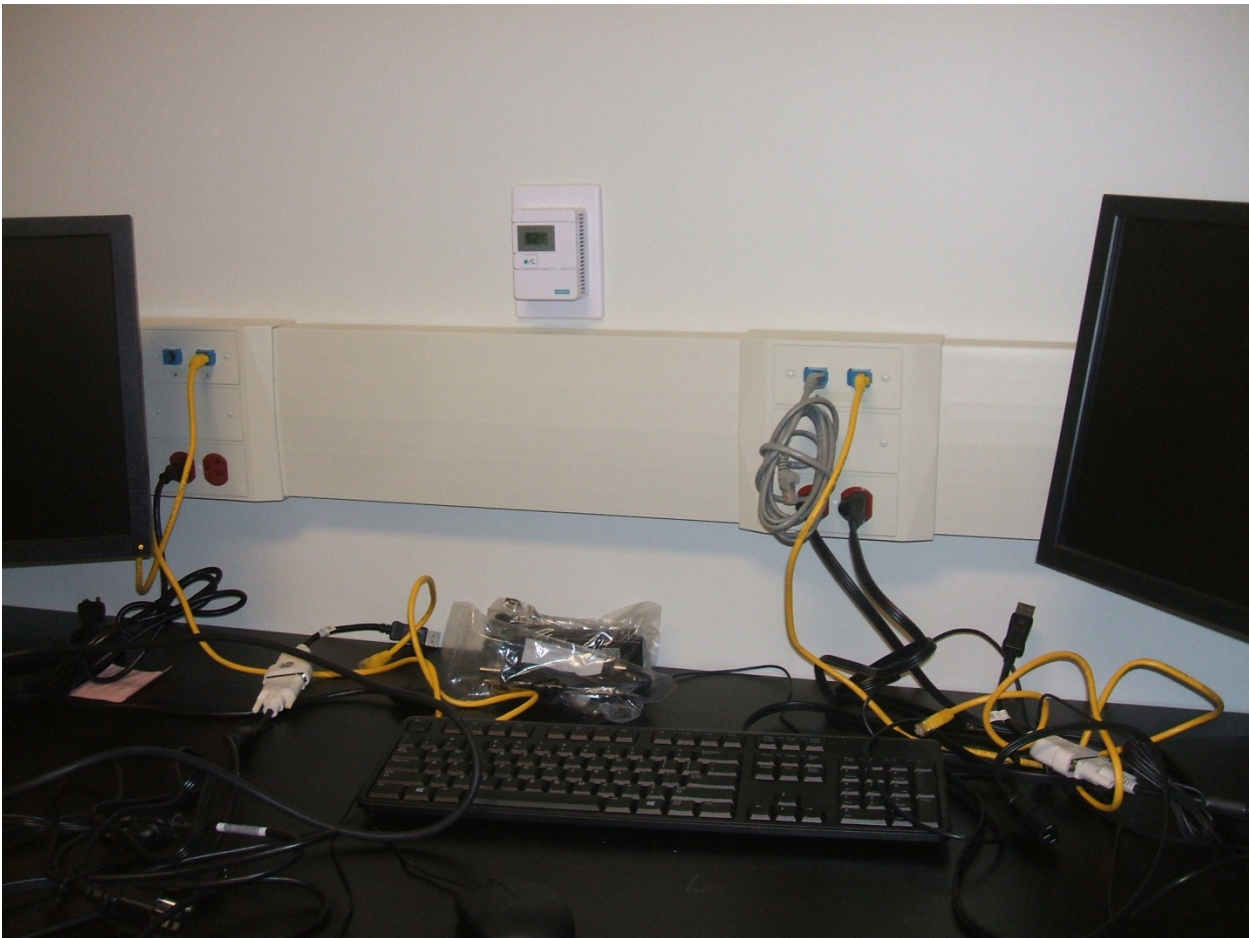
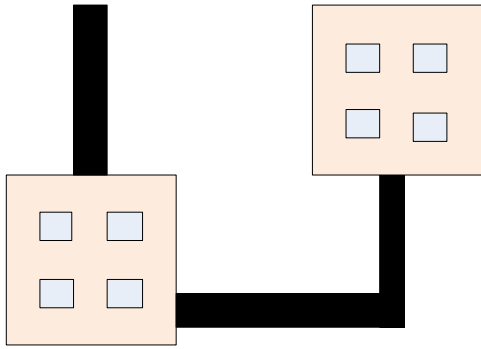


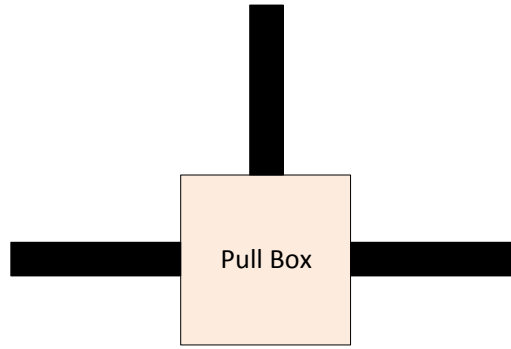
Figure
158

Above is shown Wiremold 4000 Series surface mount raceway as it is typically used in workbench/lab situations.

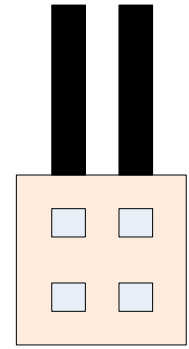
Daisy Chaining Boxes – Not Allowed



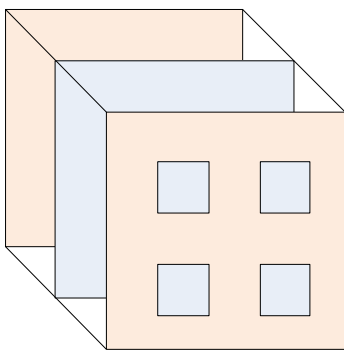
Installing modular jacks in a faceplate and extending additional cable to another back box is not allowed



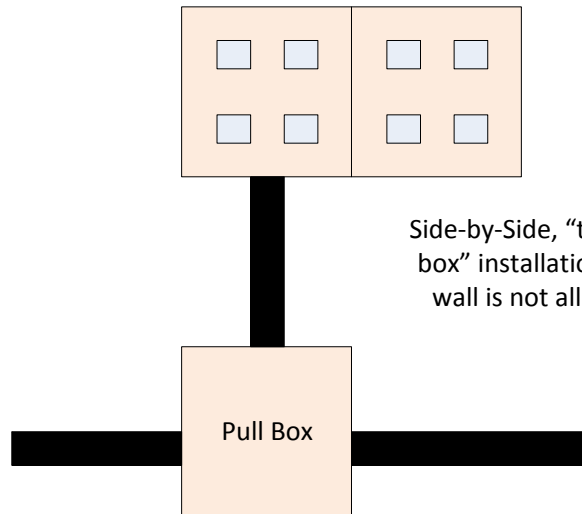
Splitting a bundle of structured cable in a Pull Box is not allowed



Using a faceplate back box as a pass through pull box for cable to other boxes is not allowed



Back-to-Back, "through box" installation, in a wall is not allowed



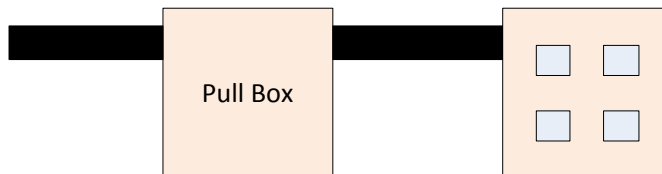
Side-by-Side, "through box" installation, in a wall is not allowed

Notes:

All conduit to be no smaller than 1 1/4"

All back boxes to be no smaller than 4 11/16", 2 1/4" deep

Correct Wiring Method



Installing a cable through a Conduit (and if necessary a) Pull Box to a single faceplate Back Box is the only approved method of installing structured cable in a wall

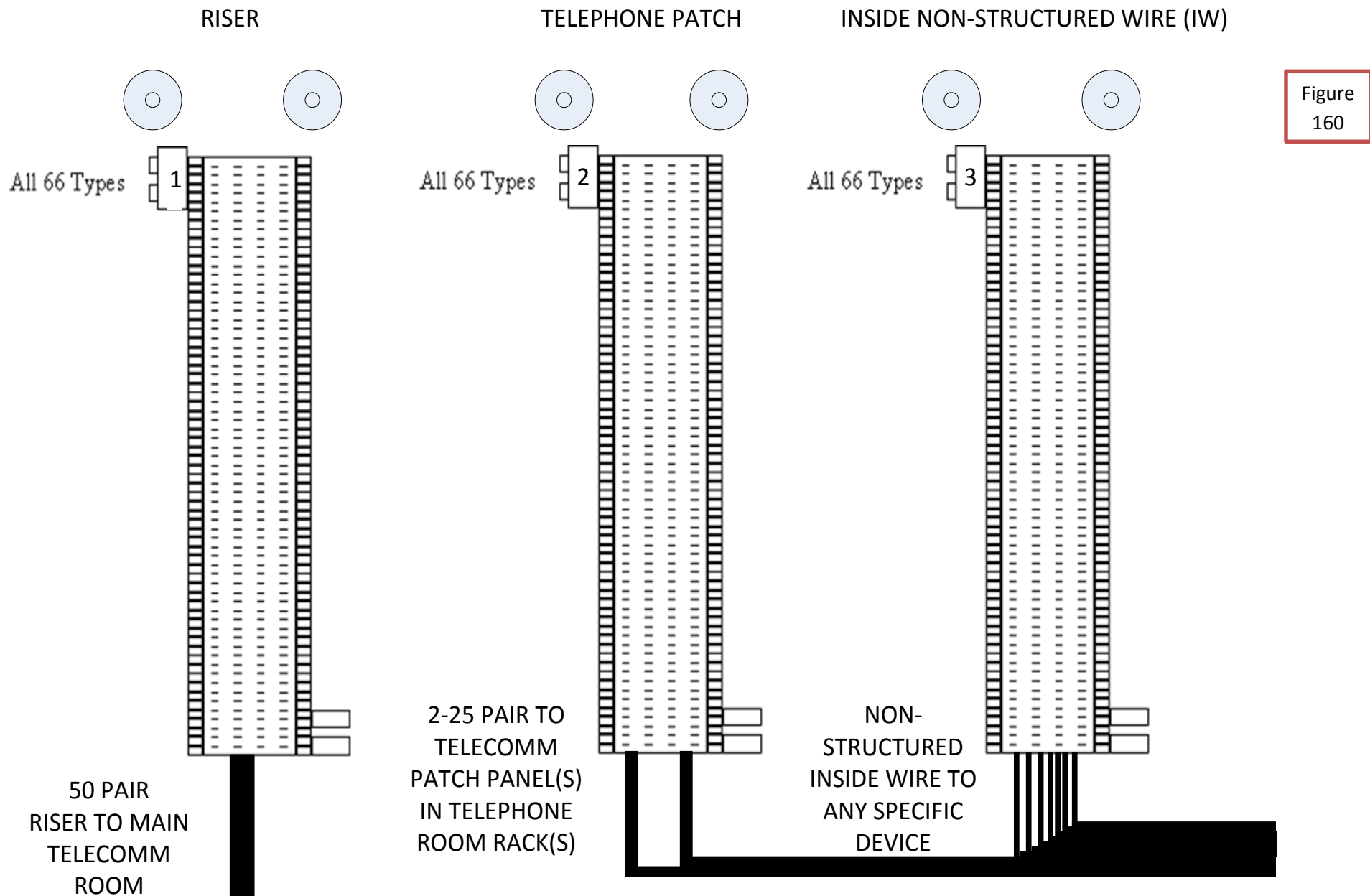
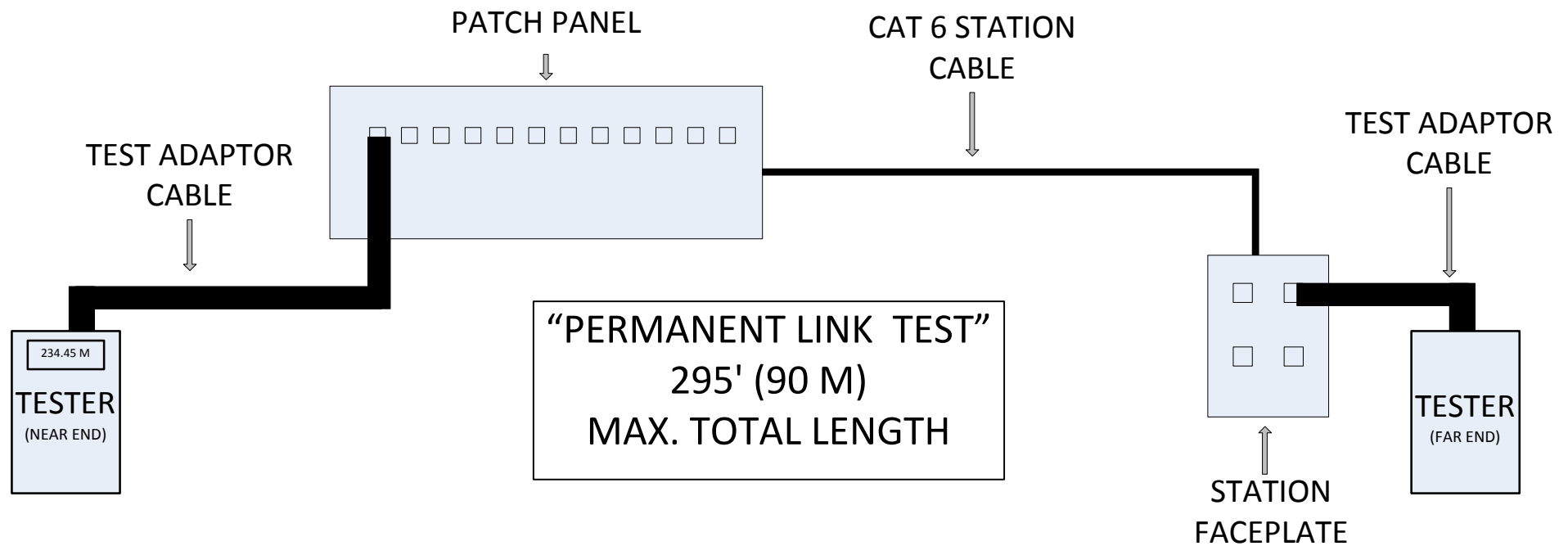
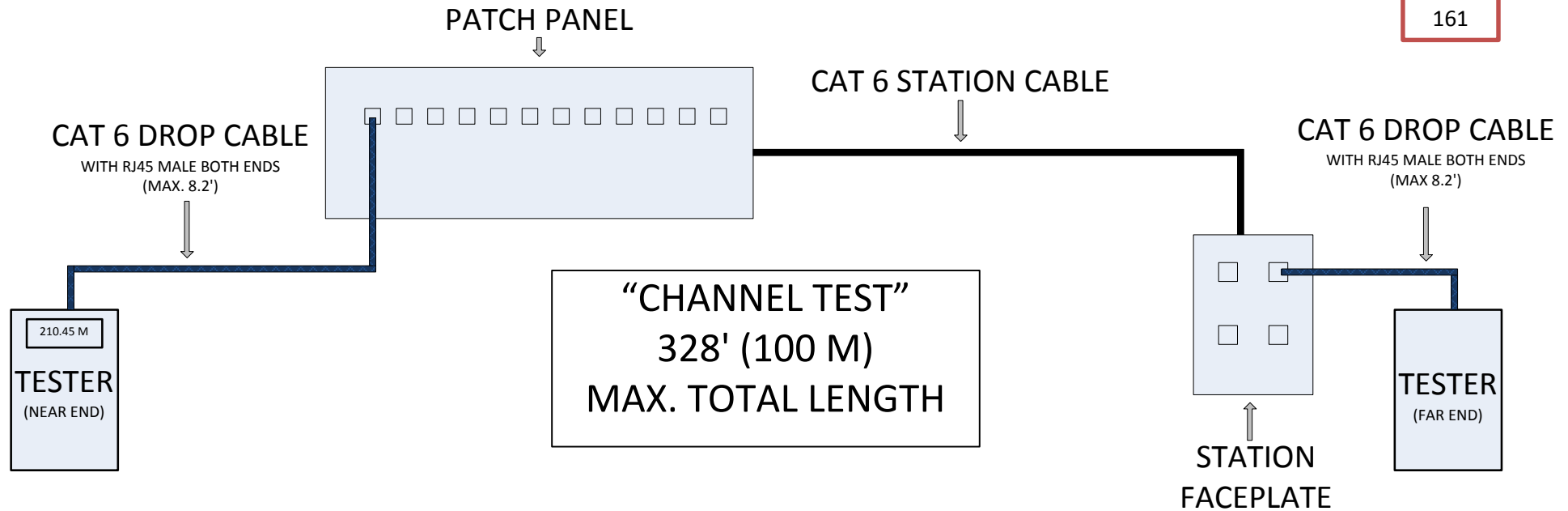


Figure
160

Telecomm back board wiring methodology, typical throughout Campus. Submit shop drawings to the ITS Telecomm group for approval before starting work.



MODE – CONDITIONING CABLE

TOTAL LENGTH – 3 METERS

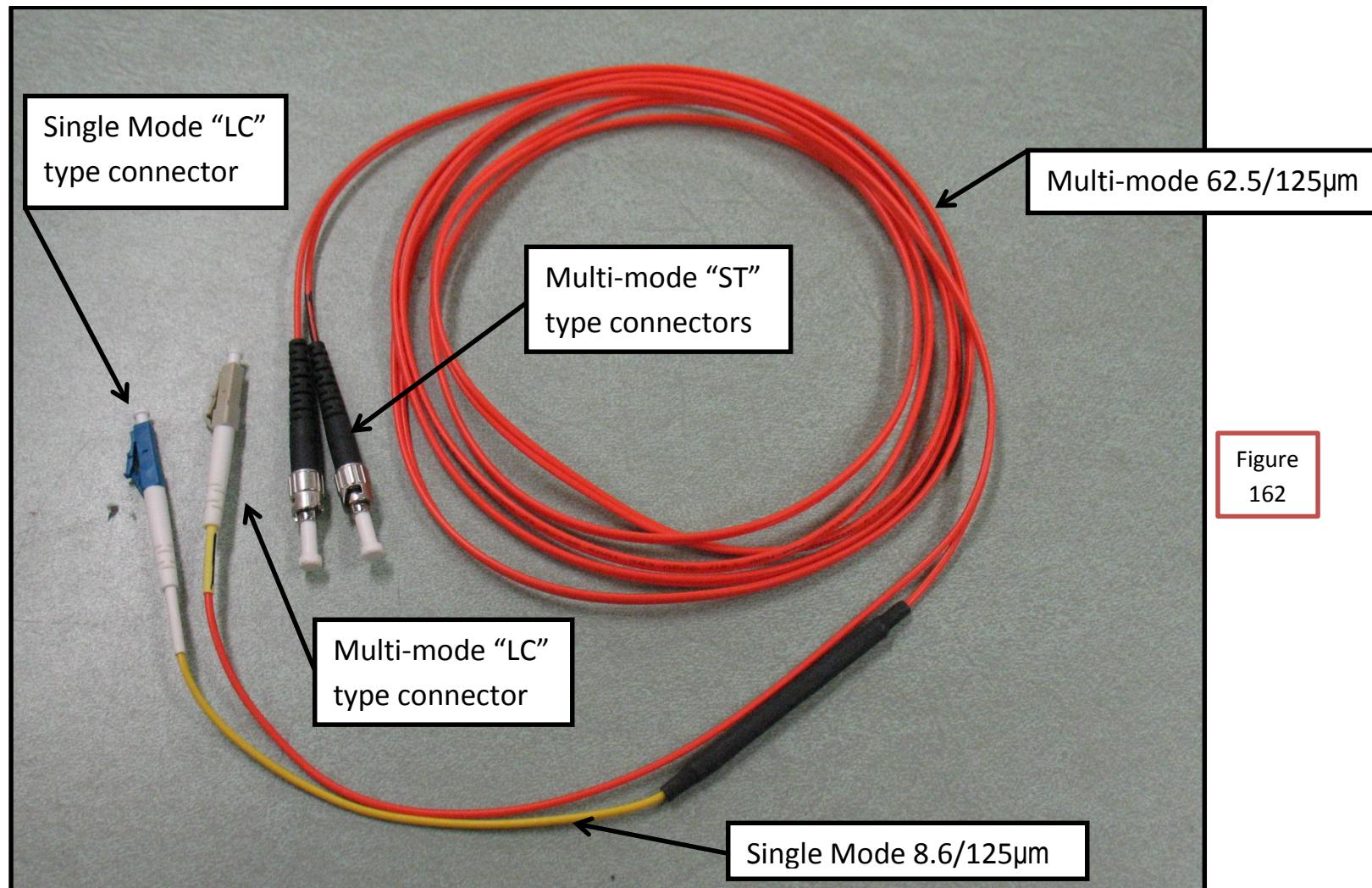
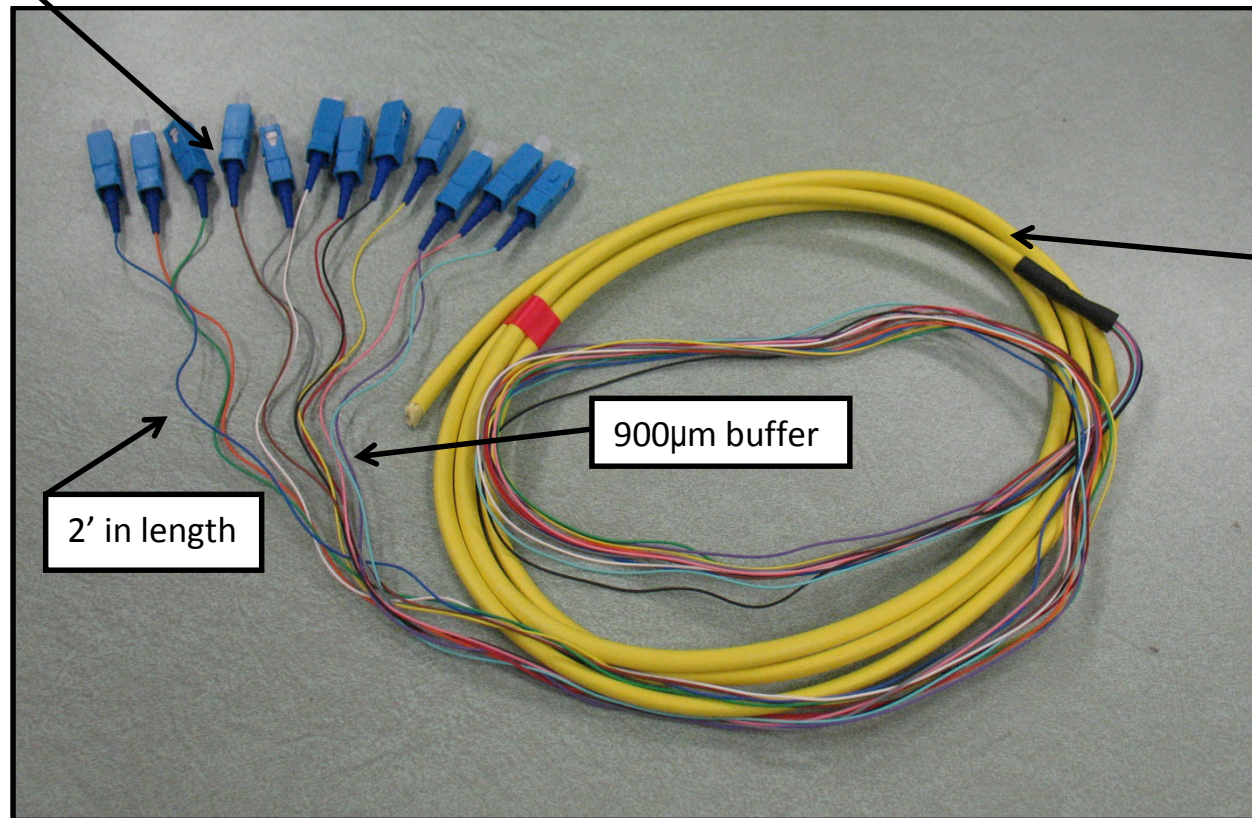


Figure
162

SINGLE-MODE PIGTAILS

Bundle of 12 colors with "SC-upc" connectors & ceramic

“SC-upc” x 12 – Blue, Orange, Green, Brown, Slate,
White, Red, Black, Yellow, Violet, Rose, Aqua



15 METERS IN LENGTH

900µm buffer

2' in length

Figure
163

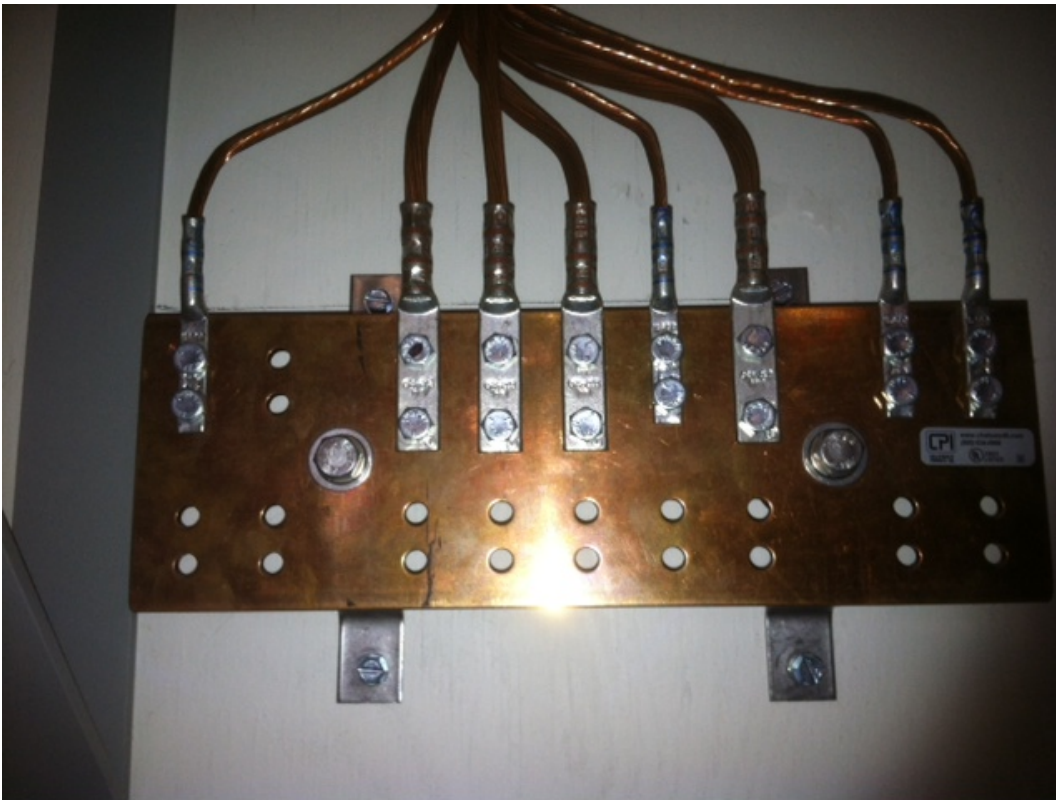


Figure
164

Telecomm Grounding Busbar (TGB) properly terminated using bare ground wire.



Figure
165

Above shows the typical vault hardware to be installed in Telecommunications Manholes (SPHs).

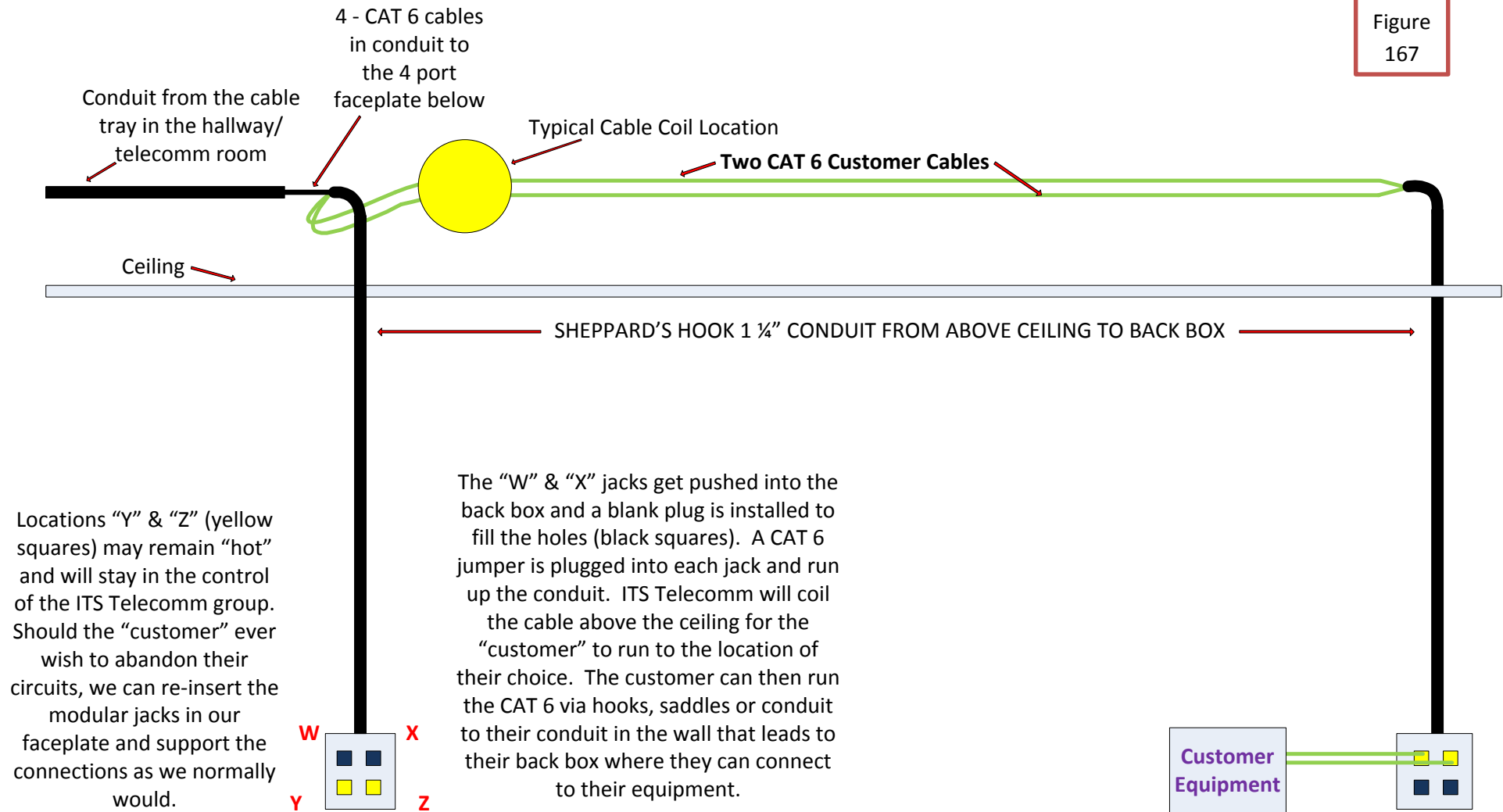


Figure
166

Shown is the proper method for installing a Cal Poly ITS Telecomm network connection in an, exposed hard ceiling, surface mounted, exposed conduit “DEMARK” environment that separates ITS Telecomm wiring/infrastructure from “customer” wiring/infrastructure. This wiring method establishes a way to quickly and easily detach customer wiring/equipment from the Structured Cabling System in an accessible location for the performance of functional testing in event of failure.

INSTALLING A CUSTOMER OBSTRUCTED NETWORK CONNECTION (DEMARK) IN A SPACE WITH HUNG CEILING, IN-WALL CONDUITS AND BACK BOXES & AN INACCESSIBLE USER FACEPLATE FOR DIRECT CONNECTION TO USER EQUIPMENT

Figure
167





Orange-White
Orange
Brown-White
Brown

Above and below show both sides of a CAT 6 modular jack terminated to the ANSI/TIS-568-C.0 Termination Standard. **The Cal Poly Campus uses the T568A wiring scheme.**

Figure
168



Blue-White
Blue
Green-White
Green